

A REVIEW OF THE STATE OF THE ART OF SOLID-STATE
CALORIC COOLING PROCESSES AT ROOM-TEMPERATURE
BEFORE 2019

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Research highlights

- ✓ A state of the art of all the developed solid-state caloric prototypes was presented.
- ✓ The devices were classified as magnetocaloric, electrocaloric, elastocaloric or barocaloric.
- ✓ The main design features and the energy performances shown were reported for each device.
- ✓ A point of view of the future perspectives is proposed in the conclusions.

A REVIEW OF THE STATE OF THE ART OF SOLID-STATE CALORIC COOLING PROCESSES AT ROOM-TEMPERATURE BEFORE 2019

A. Greco¹, C. Aprea^{2,a}, A. Maiorino^{2,c}, C. Masselli^{2,b,*}

¹ Department of Industrial Engineering, University of Naples Federico II, P.le Tecchio 80, 80125, Napoli, Italy

² Department of Industrial Engineering, University of Salerno, Via Giovanni Paolo II 132, 84084, Fisciano (SA), Italy

^a Member of IIR-IIF Commission E2

^b Member of IIR-IIF Commission B2

^c Member of IIR-IIF

* Corresponding author: e-mail cmasselli@unisa.it

Abstract

The solid-state processes for cooling and heat pump technologies are based on the well-known caloric effects (magneto-, electro-, elasto- and baro-caloric). With an interest arisen in 1976, year of the development of the first room-temperature magnetic refrigerator, up to a few years ago, magnetocaloric was among the most investigated caloric cooling techniques and it was considered among the main solid-state alternatives to vapor-compression cooling and heat-pumping. During such period, a remarkable number of prototypes of magnetic refrigerators or heat pumps was built. Recently, the attention toward all the four caloric effects has recently grown; therefore, an increase in projects on solid-state prototype developing (not only on magnetocaloric) was observed. The intention of this paper is to offer a state-of-the-art of all the solid-state prototypes for cooling and heat pumps processes, devoted to room temperature operations, developed before the year 2019.

Keywords: *Caloric refrigerator; Caloric prototype; Magnetocaloric; Electrocaloric; Elastocaloric; Barocaloric.*

NOMENCLATURE

Roman symbols

B	magnetic field induction, T
C	specific heat, J kg ⁻¹ K ⁻¹
E	electric field, V m ⁻¹
f	frequency, Hz
H	magnetic field, A m ⁻¹

M	magnetization, A m ⁻¹
$\dot{m}$	mass flow rate, g s ⁻¹
P	polarization, C m ⁻²
p	pressure, Pa
$\dot{Q}$	power, W
s	entropy, J K ⁻¹ kg
T	temperature, K
V	volume, m ³
$\dot{V}$	volumetric flow rate, l min ⁻¹
V*	dimensionless volumetric flow rate
X	conjugate field
Y	applied driving field

Greek symbols

Δ	finite difference
ε	elongation, %
μ	permeability, H m ⁻¹
ϕ	utilization factor
σ	stress, Pa

Subscripts

0	minimum
1	maximum
ad	adiabatic
H	hot reservoir
L	load
o	vacuum
ref	refrigerant
span	span

Acronyms

ACR	Active Caloric Regenerative
AER	Active Electrocaloric Regenerator
AeR	Active elastocaloric Regenerator
AMR	Active Magnetic Regenerator
BCE	BaroCaloric Effect
COP	Coefficient Of Performance
ECE	ElectroCaloric Effect
eCE	elastoCaloric Effect
ECM	ElectroCaloric Material
eCM	elastoCaloric Material
HTF	Heat Transfer Fluid
MCE	MagnetoCaloric Effect
MCM	MagnetoCaloric Material
tck	thickness
TD	Thermal Diodes

1. Introduction

Until a few years ago magnetocaloric was among the most investigated caloric cooling techniques and it was considered the only worthy of investigation solid-state alternative to vapor compression cooling and heat pumping (Brück et al., 2003). The real potential shown, both from environmentally-friendly and energy point of view, pushed towards the birth of a community of scientists and researchers devoted to a common goal: make magnetocaloric refrigeration the technology of tomorrow. With the interest to plan the research directions and to build an interactive working group, in 2004 the International Institute of Refrigeration (IIR/IIF) instituted the “Working Party on Magnetic Refrigeration at Room Temperature” and afterward, in 2005 “Thermag, the first international conference on magnetic refrigeration at room temperature”, took place in Montreux (Switzerland). Since then, eight Thermag conferences have been organized, approximately every two-years (with exception of the biennium 2009-2010 during which it took place twice - see Table 1), with the purpose to give a periodic overview of the current developments on magnetocaloric materials and prototypes. “Thermag” after “Thermag”, the number of works presented is constantly growing but, currently, together with this trend, increasing is also the interest toward the other caloric effects (electro-, elasto- and

baro-caloric). A relevant number of studies focused on materials and investigations on caloric effects different from magnetocaloric one, was presented during the last Thermag (VIII), held in Darmstadt on 16-20 September 2018; since then the conference got the novel sub-title: of “International Conference on Caloric Cooling”. This data reflects the directions towards which the scientific community, born as magnetocaloric, is now moving: in the last few years the attention on solid-state cooling was progressively extended from magnetocaloric to all the four main room temperature caloric effects applicable to refrigeration and heat pumps (Kitanovski et al., 2015a; Aprea et al., 2018a).

The purpose of this review paper is to offer a report and to describe the progress and the goals achieved by the solid-state caloric-effects community, providing a state of the art of all refrigerator and heat pump prototypes on solid-state cooling based on caloric effects for room temperature applications. With reference to the magnetocaloric ones, the emphasis will be placed on the prototypes developed after the year 2010, and therefore not mentioned by Yu et al. in their review (Yu et al., 2010).

TABLE 1

2. Solid-state refrigeration and heat pumping: principle of operations

2.1 Caloric effect phenomenon

Caloric effect is defined as all the phenomena responsible of a temperature change in a caloric material following the application of an external field which varies its intensity adiabatically. If the process is isothermal, an entropy variation in the material itself is detected. These two quantities constitute a measure of the caloric effect and they can be defined, respectively, by the following equations:

$$\Delta T_{ad} = - \int_{Y_0}^{Y_1} \frac{T}{C} \left(\frac{\partial X}{\partial T} \right)_Y dY \quad (1)$$

$$\Delta S = \int_{Y_0}^{Y_1} \left(\frac{\partial X}{\partial T} \right)_Y dY \quad (2)$$

the Y and X represent the following in each caloric effect: the applied and the conjugate ones. If one remains in the generality of X and Y, it is possible to speak of caloric effect as a macro-concept. The particularization of the above fields gives rise to specific caloric effects such as:

- magneto-, when magnetic (H) and magnetization (M) are the driving and the conjugate fields (Phan & Yu, 2007);
- electro-, where electric field (E) and polarization (P) represent Y and X, respectively;
- elasto-, driven by the mechanical stress (σ) which generates the strain (ϵ);
- baro-, in which a volume variation occurs as the conjugate result of an applied pressure field (note that for BCE case equations (1) and (2) have the signs reversed) (Moya et al., 2014).

Applying an external field in adiabatic conditions leads to temperature increase of the caloric material and it corresponds to a change in magnetization, polarization, strain and pressure if, respectively, a MCE, ECE, eCE or BCE occurs. On the contrary, decreasing adiabatically the intensity of Y and, therefore,

demagnetizing (MCE), depolarizing (ECE), outstretching (eCE) or expanding (BCE), the temperature of the caloric material decreases. To employ the caloric effect for refrigeration and heat pumping purposes and therefore to transfer heat from an environment to another, a thermodynamic cycle would be needed to do refrigeration and heat pumping continuously

2.2 Active caloric thermodynamical cycle for refrigeration and heat pump

Many full and complete reviews focusing on the most suitable thermodynamical cycles employable for solid-state refrigeration and heat pumps were published in the last two decades: most of them focused on magnetocaloric (Gómez et al., 2013a; Kitanovski et al. 2015b) but these concepts can be extended, similarly, to the whole caloric cooling and heat pumping. Therefore, more recently, some reviews on the thermodynamics of electrocaloric (Ožbolt et al., 2014) and elastocaloric (Qian et al., 2016a) systems were published. For barocaloric technology, currently, there are no comprehensive reviews, since this area is still at early stage of research. Basing on such know-how, the most efficient cycle is Active Caloric Regenerative (ACR) cycle, since it is the most effective from the point of view of developing the largest temperature gradients. ACR is composed of four processes applied to a caloric material which plays the double role of refrigerant and regenerator. It is a solid structure, made by the caloric material, crossed by a Heat Transfer Fluid (HTF) that moves in both directions, among a cold and a hot heat exchanger (Shah & Sekulic, 2003). An external field applied to the materials varies its intensity.

Figure 1 shows the four processes forming the Brayton-based Active Caloric Regenerative cycle for: (a) refrigeration and (b) heat pumping operation-modes. In (I) the intensity of the driving field grows adiabatically, provoking a temperature increase of in the material due to caloric effect. Step (II) realizes the heat transfer from cold-to-hot heat exchanger though the HTF flowing. Subsequently, the adiabatic decreasing of the intensity of the external field, that occurs in (III), provokes a decrease of temperature into the caloric material, regulated by a caloric effect. The hot-to-cold flow in step (IV) cools down the HTF that, going into the cold heat exchanger, subtracts heat and realizes the useful effect of the ACR, if it is applied for refrigeration. Dually, for heat pumping operation mode, the desired effect of the ACR cycle happens in step (II), when the fluid heats up because of cold-to-hot crossing and, reaching the hot heat exchanger, it adds heat to the environment to be warmed.

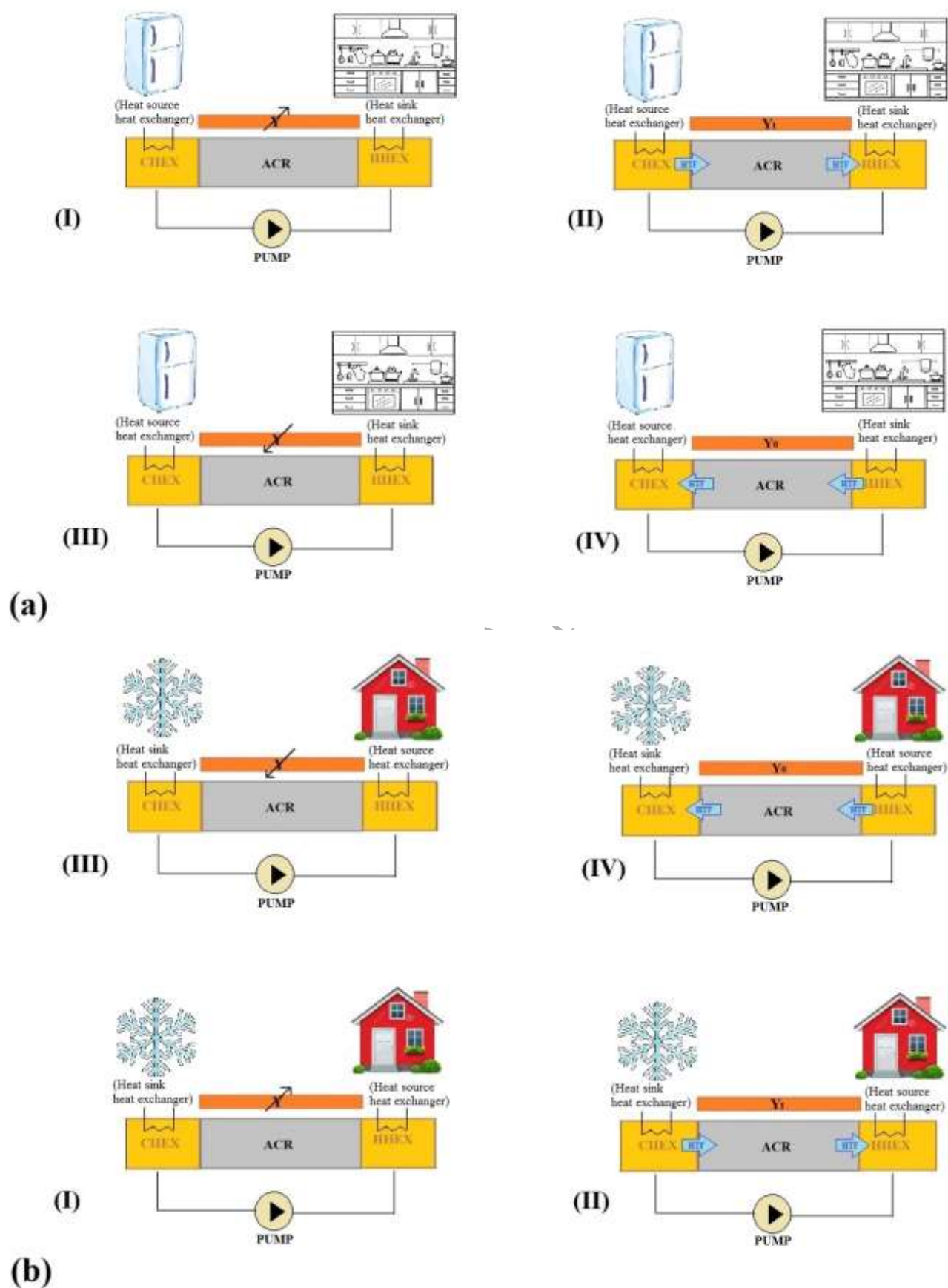


Figure 1. The four steps of Active Caloric Regenerative cycle: I) adiabatic field-growing; II) cold-to-hot flowing; III) adiabatic field-falling; IV) hot-to-cold flowing for (a) refrigeration; (b) heat pumping operation modes.

An alternative to the active regenerative process based on convection between solid and a fluid, is constituted by the application of Thermal Diodes (TD) in a solid-to-solid caloric thermodynamical cycle to manipulate and control the heat transfer direction. The thermodynamical of this cycle is well reported by Kitanovski et al. in their review on caloric refrigeration and heat pump technologies (Kitanovski et al., 2015b). However, the thermal diodes processes are not described in detail in the present paper since the purpose of our review is the classification of the prototypes of solid-state cooling processes, based mainly on Active Caloric Regenerative refrigeration-heat pumping cycle.

3. Classification of the magnetocaloric prototypes operating at room temperature

3.1 Earlier than the year 2010

All the magnetocaloric prototypes presented before the year 2010 are classified and described exhaustively by Yu et al. in their review (Yu et al., 2010): they provided a detailed historical review of the progress made by magnetic refrigeration toward the development of room temperature prototypes starting from the dawn of the technology when, in 1976, Brown presented the first magnetic heat-pump working at room-temperature (Brown et al., 1976). The machine operated with superconductive magnets ($\Delta B=7T$) and with parallel-plates of gadolinium crossed by a mixture of 20% ethylic alcohol – 80% water and it was based on Stirling cycle with passive regeneration. The latter feature was not optimal for performing a thermodynamical cycle (Barclay, 1994). In 1978 Steyert proposed (Steyert, 1978) a magnetocaloric device like the one by Brown (Brown et al., 1976) but operating with Active Magnetic Regenerator (AMR). The introduction of AMR concept marked a turning point, since most of the subsequent projects on magnetocaloric devices based on it. In their review (Yu et al., 2010) Yu et al. classified the magnetocaloric devices dividing them in a first and a second generation of room temperature magnetic refrigerators and heat pumps. The first generation identified all the room-temperature-operating devices that presented the common points of: employing superconducting magnets and working at low-frequency of operation. The transition to the second generation prototypes occurred when the Astronautic Corporation of America presented the first rotating magnetocaloric refrigeration, mounting permanent magnets (Zimm et al., 2003). The application of rotating permanent magnets allowed to reach higher frequencies of operation. It gives the birth to the second generation of magnetocaloric prototypes, showing the common points of being rotary and operating with higher frequencies. Yu et al. in their review, summarized all the developed prototypes in a table (Yu et al., 2010), reported here as Figure 2, that will be our starting point for the prosecution of the magnetocaloric-prototype classification.

Name and location	Date of announce-ment	Type and max. frequency (Hz)	Max. cooling Power (W)	Max. ΔT (K)	Max. magnetic field strength (T)	Magnetocaloric regenerator material
1 Brown (1976, 1978) NASA, USA	1976	Reciprocating, Styrling/	/	47 (80)	Supercond. 7	1 mm Gd plates, 20% ethyl alcohol- water
2 Kirtel and Dacuz (1987) Idaho National Engineering Laboratory, USA	1987	Rotating type, Ericsson cycle 0.5 Hz	/	11	Permanent 0.9 T	125 Gd plates 76 μ m, spacing 127 μ m, water
3 Green et al. (1993) US NAVY, USA	1990	Reciprocating, 0.02 Hz	/	24	Supercond. 7	Gd-Tb metal ribbons, Nitrogen gas
4 Zimm et al. (1998) AMES Ast. Corp. A, USA	1997	Reciprocating	600	38	Supercond. 5	Gd spheres 0.15–0.3 mm
5 Bohigas et al. (2000) Mater. Science Institute, Barcelona, Spain	2000	Rotary 0.8 Hz	/	5	Permanent 0.9	Gd as ribbon on plastic disk, olive oil
6 Okamura et al. (2005) Tokyo Inst. Tech Chubu Electric Power, Japan	2000	Reciprocating	100	21	Supercond. 4	Gd spheres
7 Rowe and Bartley (2002) Univ. Victoria, Victoria, Canada	2001	Reciprocating	2	14	Supercond. 2	Gd & Gd _{1-x} Th _x layered beds of grains, Helium
8 Zimm et al. (2003, 2005, 2006), Astronautics Corp. America, Madison, Wisconsin, USA	2001	Rotary 4 Hz	50	25	Permanent 1.5	Gd, GdEr spheres, 0.25–0.5 mm, water with inhibitors
9 Blumenfeld et al. (2002) Los Alamos Natl. Lab. New Mexico, USA	2002	Charging- discharging a coil, no moving parts	3	15	Supercond. 1.7	Gd powder.
10 Lu et al. (2005) Sichuan Inst.Tech./Nanjing University, China	2002	Reciprocating	/	23	Permanent 1.4	Gd spheres, 0.2 mm, Gd ₂ Si _{1.88} Ge _{1.00} Gd _{0.03} powder
11 Okamura et al. (2005) Tok Inst.Tech. Chubu Electric Japan	2003	Rotary (rotation of magnets) 0.4 Hz	60	8	Permanent 0.77	Gd _{1-x} Dy _x layered alloys, spheres 0.6 mm, 1 kg, water
12 Clot et al. (2003) Lab. d'Electrotechnique Grenoble, France	2003	Reciprocating 0.5 Hz	8.8	4	Permanent 0.8	Gd foils, 1 mm with gaps for water, 0.1 mm
13 Richard et al. (2004) Univ. Quebec, Trois Rivières, Canada	2004	Reciprocating	2	14	Supercond. 2	Gd-R alloys layered beds of grains
14 Rowe et al. (2005, 2006) Univ. Victoria Canada	2004	Reciprocating 1 Hz	10	50	Supercond. 2	Gd-R layered alloys, Helium
15 Shir et al. (2005) George Washington Univ., Ashburn, Virginia, USA	2005	Reciprocating	/	5	Permanent 2	Gd powder
16 Okamura et al. (2007) Tokyo I. Tech., Chubu Electric Power, Japan	2005	Rotary	110	10	Permanent 0.77	Gd alloys MnAsSb alloys, and Gd packed beds Water
17 Vasilie and Müller (2005, 2006) INSA, Cooltech Applications Strasbourg, France	2005	Rotary	360	14	Permanent 2.4	Gd plates, water
18 Yu et al. (2005) School of Energy and Power Engineering, Xi'an Jiaotong University Xi'an, China	2005	Reciprocating	18.7	3	Electro-magnet (water cooled) 2.18	Gd 0.3 mm and Gd ₂ Si ₂ Ge ₂ particles, water
19 Kawamura et al. (2005) Graduate School of Engineering, Hokkaido University, Japan	2005	Reciprocating ~0.2 Hz	/	10	Permanent 1.0	Gd, 63% vol. fraction, distilled water
20 Yao et al. (2006) Technical Institute of Physics and Chemistry, Beijing, China	2006	Reciprocating 1 Hz	51	42	Permanent 1.5	Gd particles, helium
21 Egolf et al. (2006a-c) University of Applied Sc. of Western Switzerland, Switzerland	2006	Rotary	-	-	Permanent 0.8	Gd particles, air
22 Okamura et al. (2005, 2007) Tokyo Inst. Tech., Chubu Electric Power, Japan	2006	Rotary 0.5 Hz	560	8	Permanent 1.1	Gd foils, water
23 Huang et al. (2006, 2007) Baotou Research Institute of Rare Earth, China	2006	Reciprocating	50	18	Permanent 1.5	Gd 750 g and LaFe _{10.9} Co _{0.78} Si _{1.08} Bi _{0.2} 200 g, alkaline water solution, Ph = 10
24 Zimm et al. (2007) Astronautics Corp. America Madison, Wisconsin, USA	2007	Rotary 4 Hz	220	12	Permanent 1.5	Gd plates, water
25 Turn and Rowe (2007) Univ. Victoria Victoria, Canada	2007	Rotary 4 Hz	/	13.2	Permanent 1.47	Gd flakes, 0.6 mm, water
26 Buchelnikov et al. (2007) Chelyabinsk State University Russia	2007	Rotary 10 Hz	40	/	Permanent 1	Gd, NiMnGa alloys
27 Petersen et al. (2007) Risø National Laboratory, Denmark	2007	Reciprocating <1 Hz	/	8.7	Permanent 1.2	Gd plates, 1 mm
28 Muller et al. (2007) INSA, Cooltech Applications Strasbourg, France	2007	Rotary magnet	/	/	Permanent 1.3 T	Gd, water
29 Chen et al. (2007) School of Materials Science and Engineering, Sichuan University Chengdu, China	2007	Rotary magnet 0.7 Hz	40	11.5	Permanent 1.5	Gd particles, 0.5 mm, 1 kg, water
30 Nakamura et al. (2008) Graduate School of Engineering, Hokkaido University in Japan	2008	Reciprocating	/	/	Permanent 2.0	Gd particles, 33.4 g, water or air
31 Hirano et al. (2009) Hokkaido Industrial Research Institute, Japan	2009	Reciprocating	/	~2	Permanent 2.3	LaFeSi-type material, air
32 Zheng et al. (2009) South China University of Technology, Chian	2009	Reciprocating	/	/	Permanent 1.5	Gd
33 Bour et al. (2009), Cooltech Applications, France	2009	Reciprocating 1 Hz	/	16.1	Permanent 1.1	Gd plates, 0.6 mm, water
34 Coelho et al. (2009) Universidade Estadual de Campinas, Brazil	2009	Rotary 0.5 Hz	/	11	Electro-magn. 2.3	Gd pins, ethyl alcohol
35 Dupuis et al. (2009) Grenoble Electrical Engineering Laboratory, France	2009	Reciprocating 1 Hz	/	7.8	Permanent 0.8	Gd sheets, 1 mm
36 Kim and Jeong (2009) Korea Advanced Institute of Science and Technology, Korea	2009	Reciprocating 1 Hz	/	16	Permanent 1.58	Gd particles, 325–500 μ m, Helium
37 Fryds et al. (2009) Risø DTU, Denmark	2009	Reciprocating	/	/	/	La _{0.9} Ca _{0.1} – _{0.9} Br _{0.9} MnO ₃ plates, water and ethanol mixture
38 Sari et al. (2009) IGT in University of Applied Sciences of Western Switzerland, Switzerland	2009	Reciprocating	/	/	Permanent 2.0	LaFe ₁₁ Co _{0.5} Si _{0.1} and Gd, air
39 Tagliafico et al. (2009) University of Genoa, Genoa, Italy	2009	Reciprocating <0.2 Hz	/	/	Permanent 1.5	Gd, 300 μ m, water
40 Turn and Rowe (2009) University of Victoria, Victoria, Canada	2009	Rotary 4 Hz	50	29	Permanent 1.4	Gd spheres, 300 μ m
41 Tušek et al. (2009) University of Ljubljana, Slovenia	2009	Rotary 4 Hz	/	~7	Permanent 0.96	Gd plates, 0.3 mm, ~600 g, distilled water

Figure 2. From Yu et al., 2010, a list of all the room temperature magnetocaloric prototypes earlier than 2010.

3.2 From the year 2010 up to the year 2019

In the last decade a huge increment of developed prototypes was registered. The number of research groups working on magnetocaloric, and then more generally on caloric, is growing day after day.

In their review Yu et al. (Yu et al., 2010) mention the announcement in 2009, during Thermag III, by the University of Victoria, of being working on a second-generation prototype. Tura and Rowe, introduced (Tura and Rowe, 2011) their “PM-I”, displayed in Figure 3, presenting it as a rotary device mounting a magnetic system based on permanent magnets assembled in a couple of concentric nested Halbach cylinders ($\Delta B=1.5$ T); the system was acting on two Active Magnetic Regenerators crossed by a 80%water – 20%glycol mixture, as secondary fluid (Tura and Rowe, 2011). Tura and Rowe conducted the tests with regenerators filled with gadolinium both as spheres (0.3 mm) and parallel-plates (100 μm thick) and they concluded that PM-I can produce a maximum temperature span of 29 K under zero load conditions and maximum $\dot{Q}_{ref}$ of 50 W in correspondence of $\Delta T_{span} = 10$ K (Tura and Rowe, 2011).

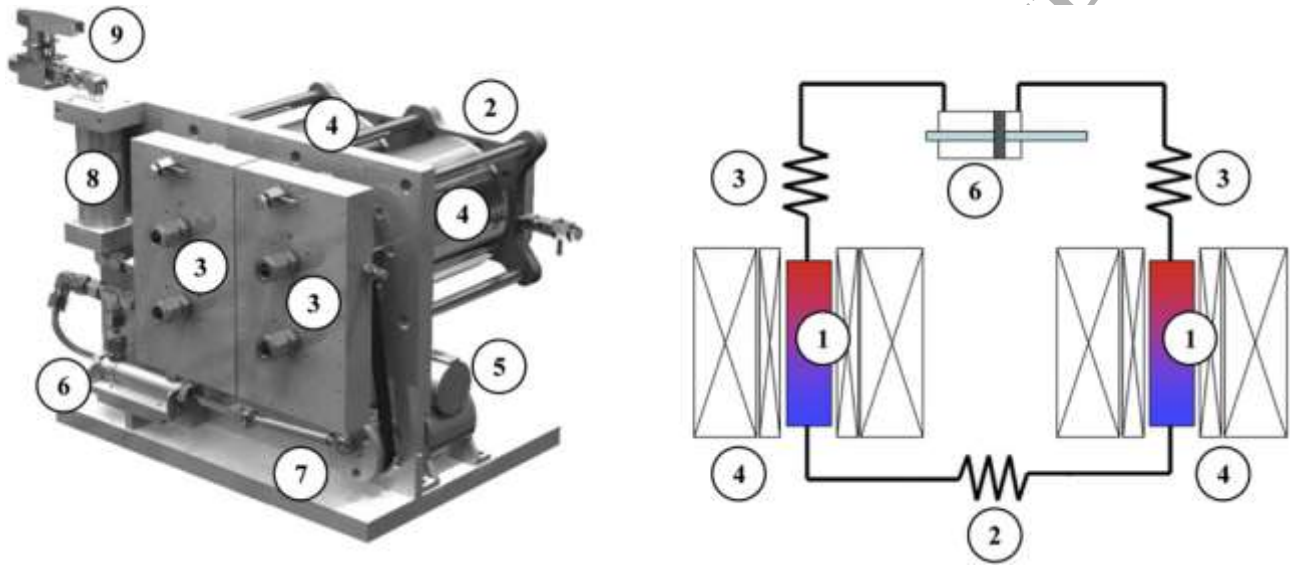


Figure 3. The first rotary prototype, based on permanent magnets, introduced by University of Victoria: (a) a photograph and (b) the hydraulic circuit. (Tura and Rowe, 2011).

In 2014 the same group presented (Arnold et al., 2014) a new rotary prototype, called PM-II and shown in Figure 4, employing three concentric nested Halbach cylinders, instead of PM-I in which there were only two, whose rotation gives $\Delta B=1.5$ T. Moreover, among the main novelties compared to the previous version (PM-I) there is a new version of the regenerator, presenting an augmented maximum volume per regenerator (57 cm^3 with respect to the PM-I 22 cm^3) to lower pressure drop and higher cooling power. The device mounts two AMRs filled with gadolinium in spherical particles (0.5 mm diameter) and crossed by 80% water – 20% glycol mixture as HTF. Arnold et al. state (Arnold et al., 2014) that PM-II can provide 33 K as maximum temperature span with no-cooling load and that maximum exergetic cooling power is around 2.5 W in correspondence of $\Delta T_{span}=15$ K.

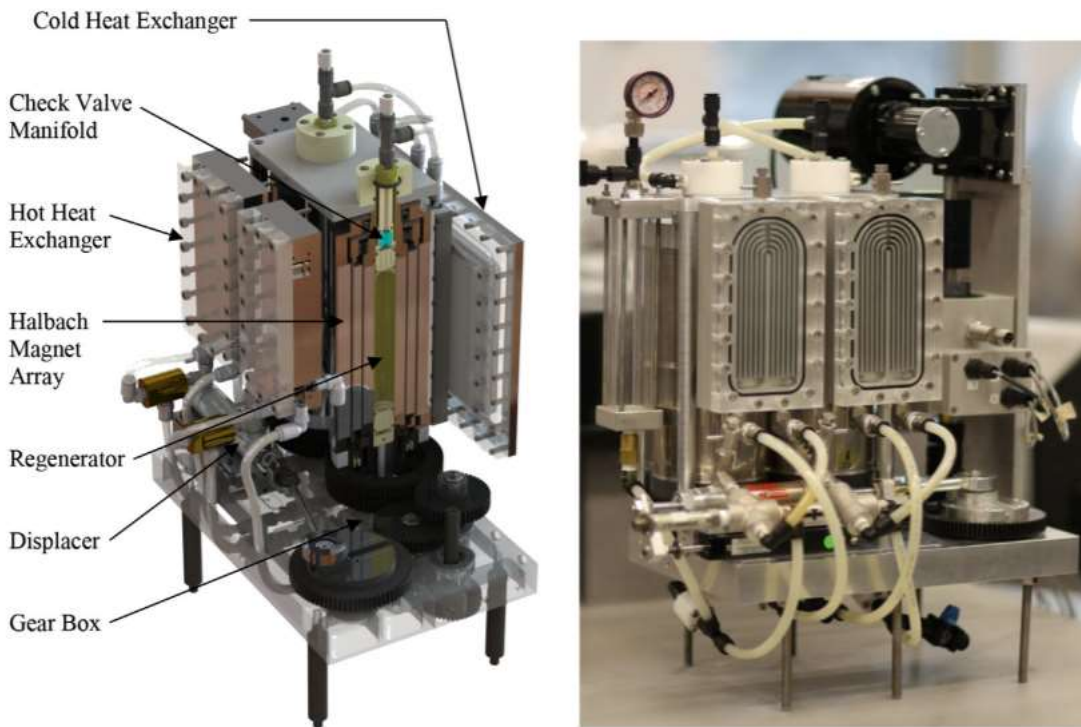


Figure 4. PM-II, the second AMR rotary prototype developed by University of Victoria (Arnold et al., 2014).

The research group of Technical University of Denmark is among the most active in the field of caloric prototype developing: in 2011 Engelbrecht et al. provide (Engelbrecht et al., 2011) to scientific community detailed experimental tests on their first prototype (whose development was announced in 2009 during Thermag III). They presented the prototype as reciprocating in which the parallel-plates regenerator is moving with respect to a magnetic system whose configuration is Halbach array ($\Delta B = 1.03$ T). Engelbrecht et al. tested (Engelbrecht et al., 2011) the device with three different MCE materials (Gd; lanthanum-based alloys of $\text{La}(\text{Fe}, \text{Co}, \text{Si})_{13}$, $(\text{La}, \text{Ca}, \text{Sr})\text{MnO}_3$) and 25% antifreeze-75% water mixture as HTF and they detected a maximum temperature span of 10.2 K under no-load conditions with gadolinium.

In 2012 the same group, in collaboration with Federal University of Santa Catarina, introduced (Engelbrecht et al., 2012) a rotary regenerator AMR device on which a stationary permanent magnet magnetic system, assembled as concentric Halbach array cylinders, give $\Delta B_{\text{max}} = 1.24$ T and $\Delta B_{\text{medium}} = 1.24$ T. The device, shown in Figure 5, mounts 24 regenerators of Gd spheres ($0.25 \div 0.8$ mm). Engelbrecht et al. asserted (Engelbrecht et al., 2012) that during the tests the device produced $\Delta T_{\text{span, max}} = 25.4$ K under zero cooling load at $f = 2$ Hz and $\dot{V} = 6.7$ l min⁻¹. Lately, the research group demonstrated (Lozano et al., 2014), by means of further tests, that the device can produce 1 kW as cooling power with $\Delta T_{\text{span}} = 0.3$ K ($f = 1.8$ Hz; $T_H = 292.3$ K; $\dot{V} = 11.4$ l min⁻¹).

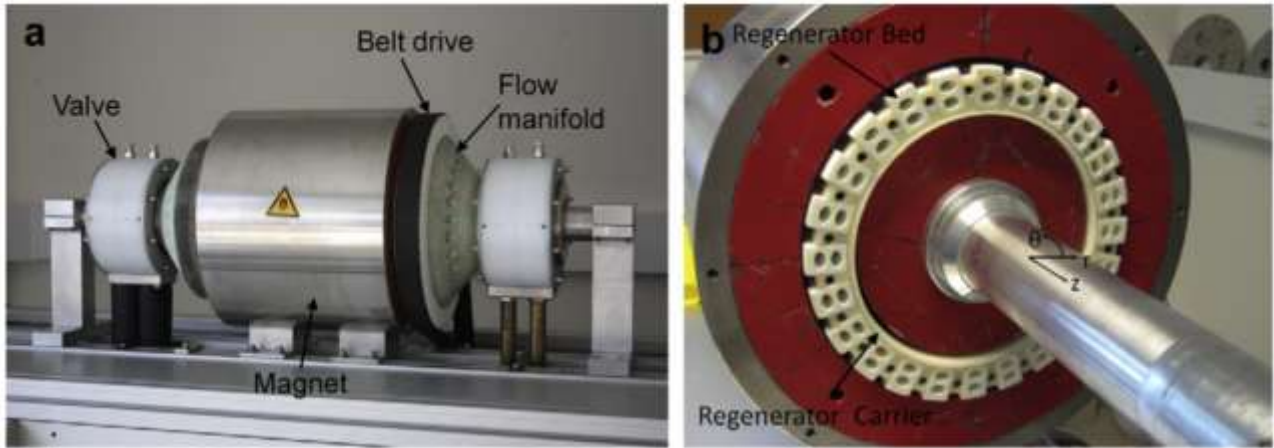


Figure 5. A rotary regenerator AMR device built at Technical University of Denmark in collaboration with Federal University of Santa Catarina: (a) the device; (b) the regenerator assembled in the device (Engelbrecht et al., 2012).

In 2014, during Thermag VI, Eriksen announced (Eriksen et al., 2014) the group of the Technical University of Denmark building a third prototype, a rotary AMR, then called MAGGIE, reported in Figure 6. The objective was producing temperature spans and cooling powers relevant for commercial refrigeration. The device well-presented and tested by Eriksen et al. (Eriksen et al., 2015) is composed by a magnetic system in Halbach assembly ($\Delta B_{\text{maximum}}=1.4$ T; $\Delta B_{\text{medium}}=1.13$ T); the AMR regenerator is filled with spheres (0.3÷0.6 mm) of Gd and three different compositions of $\text{Gd}_{100-x}\text{Y}_x$, which is crossed by a 5% glycol ethylenic - 95% demineralized water mixture. Eriksen et al. said (Eriksen et al., 2015) that a maximum COP of 3.6 was obtained under $\Delta T_{\text{span}}=7.2$ K and $\dot{Q}_{\text{ref}}=0.3$ W and that the device can operate up to 4 Hz, but maximum temperature span (20 K) was registered at $f=0.75$ Hz, under no-load. The best result that they observed (Eriksen et al., 2015) was $\text{COP}=3.1$ with $\Delta T_{\text{span}}=10.2$ K and $\dot{Q}_{\text{ref}}=102.8$ W.

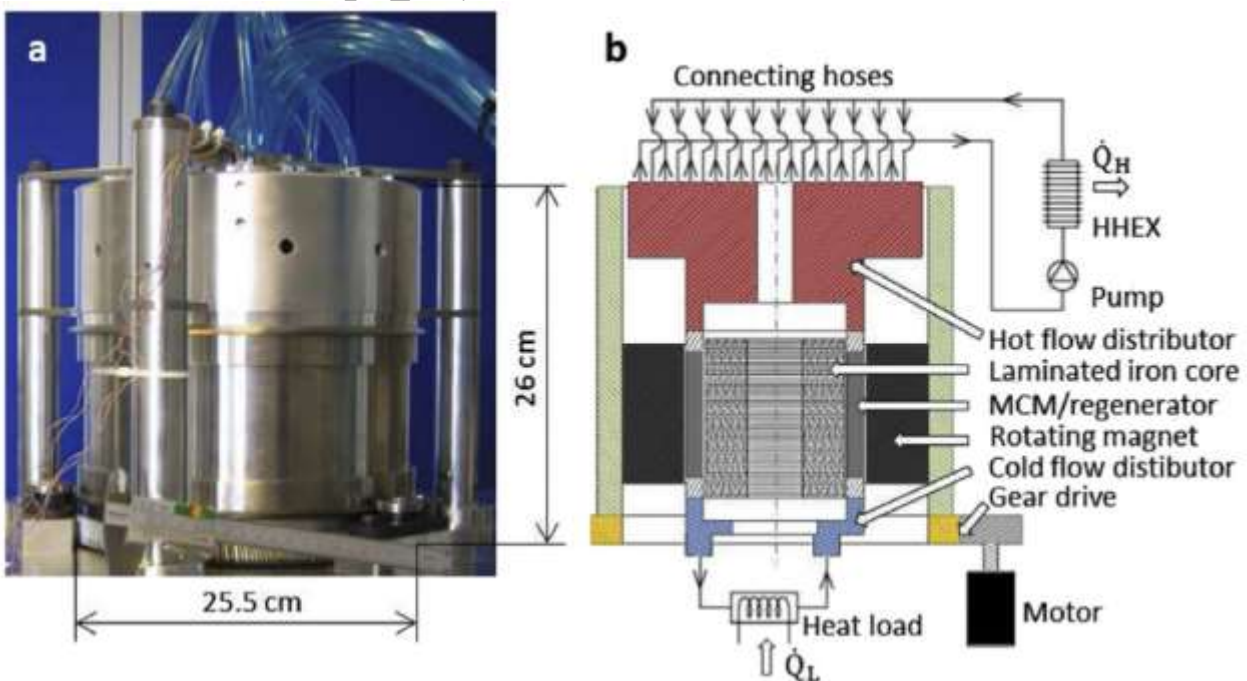


Figure 6. MAGGIE, the second rotary AMR prototype built at DTU: (a) a photograph of the device; (b) a simplified scheme showing the components (Eriksen et al., 2015).

In 2018, during the last Thermag Conference (Thermag VIII), held in Darmstadt, within the ENOVHEAT project, which brings together five universities and four industrial partners, MagQueen, a magnetocaloric heat pump was presented (Dall'Olio et al., 2018). Johra et al. state (Johra et al., 2019) that the project has the objective of realizing and investigating a prototype of an AMR heat pump for building applications. In the device, shown in Figure 7, the permanent magnets system ($B_{\max}=1.6$ T) is rotating and the AMR regenerators (13 packed-beds, each one filled with 10 layers of $\text{La}(\text{Fe,Mn,Si})_{13}\text{H}_z$ alloys) are positioned on a ring. Johra et al. said (Johra et al., 2019) that the maximum operating frequency is 2 Hz, while the flow rate can reach 2000 l h^{-1} and during initial tests they concluded that MagQueen gives 5 as maximum coefficient of performance, 1500 W as maximum heating power, in correspondence of a temperature span of 25 K.

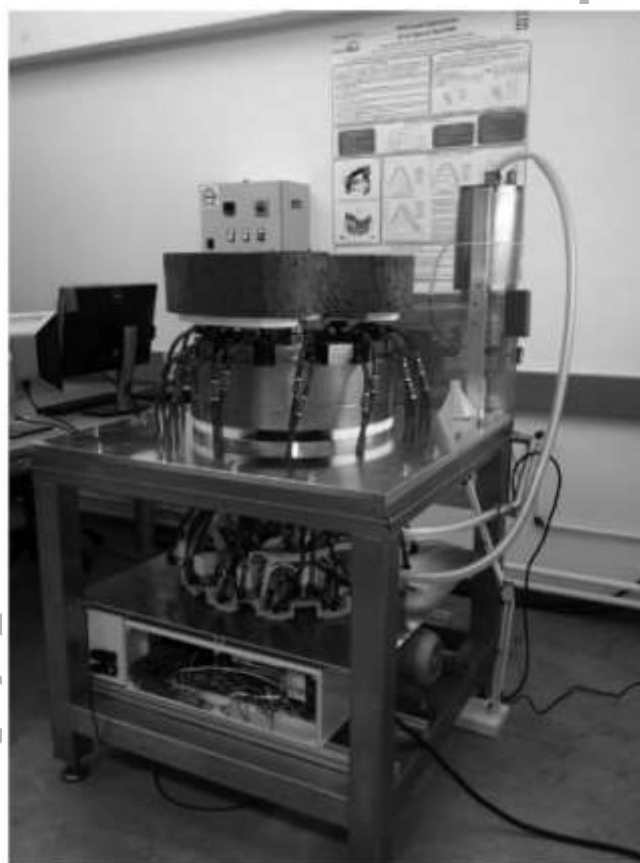


Figure 7. The ENOVHEAT project heat pump prototype (Dall'Olio et al., 2018).

In addition to the collaboration with the Danish research group, the Federal University of Santa Catarina research group was very active in developing solid-state prototypes since 2011, when Trevizoli et al. presented (Trevizoli et al., 2011) the first Brazilian alternative AMR device, mounting a magnetic group formed by Halbach permanent magnets ($B_{\max}=1.65$ T) which acts on a parallel-plates regenerator (0.85 mm thick) made of gadolinium. In the channels stacked by the parallel-plates the HTF (water) flows. Trevizoli et al. observed (Trevizoli et al., 2011) during zero-load tests, the device giving $\Delta T_{\text{span,max}}=4.45$ K with $T_H=296$

K; they detected a $\dot{Q}_{ref,max}=3.9$ W with $T_H=296$ K and 0.9 as utilization factor. During the 24th IIR International Congress of Refrigeration, in 2015, the same group announced the development of a novel rotary magnetic refrigerator (Lozano et al., 2015). Lozano et al., introduced (Lozano et al., 2015) the prototype (Figure 8) assembling a rotative magnetic system ($\Delta B=1$ T) upon 16 regenerators in which gadolinium spheres were packed in (0.425–0.6 mm diameter). During the tests Lozano et al. found (Lozano et al., 2016) the device exhibiting $\Delta T_{span,max}=12$ K ($f=1.5$ Hz and $\dot{V}=150$ l h⁻¹) in zero-load conditions and $\dot{Q}_{ref,max}=150$ W under zero-span ($f=0.8$ Hz and $\dot{V}=200$ l h⁻¹); moreover, for a thermal load of 80.4 W, under the same previous conditions, they observed the device giving 7.1 K as ΔT_{span} in correspondence of 0.54 as COP.

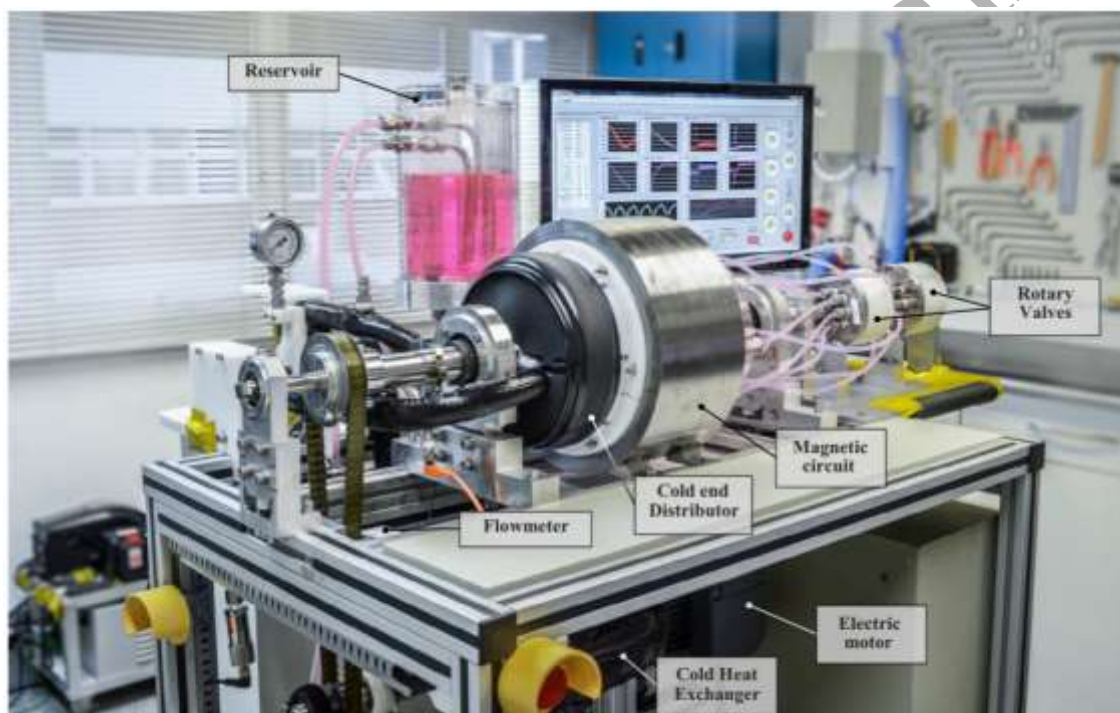


Figure 8. The rotary device built at Federal University of Santa Catarina (Lozano et al., 2016).

In 2016, the same Brazilian group, built a new active magnetic regenerator laboratory apparatus, reported in Figure 9 (Trevizoli et al., 2016): the magnetic system is made of permanent magnets nested as Halbach cylinder ($\Delta B_{max}=1.69$ T) acting alternatively on a single packed-bed regenerator (filled with gadolinium spheres) in which 80%water- 20% ethylene glycol mixture vehiculates heat fluxes. Trevizoli et al., attested (Trevizoli et al., 2016) that the prototype can reach a maximum cooling power of 53.7 W (with $\phi=0.77$ and $f=1$ Hz) and a maximum temperature span of 30 K under zero cooling-load, whereas the highest COP detected was 4.6 (with $f=0.25$ Hz and $\phi=0.77$).

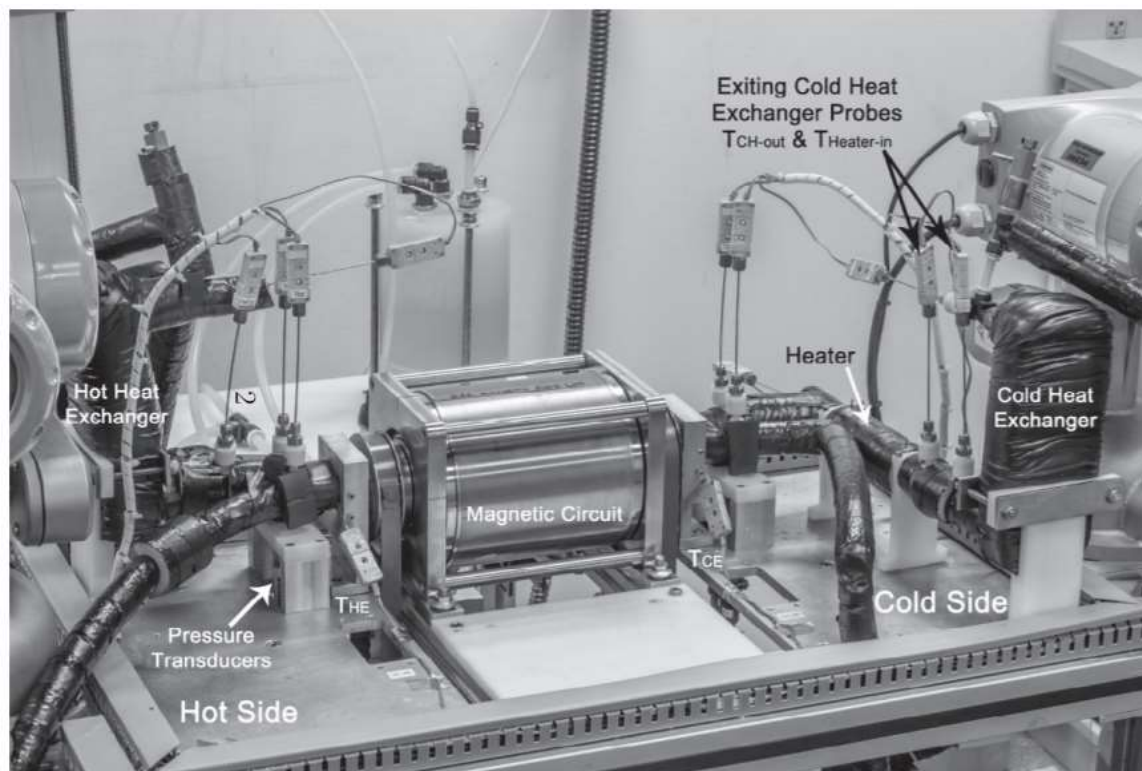


Figure 9. The active magnetic regenerator laboratory apparatus built by Federal University of Santa Catarina (Trevizoli et al., 2016).

Recent updates about the activities of Federal University of Santa Catarina see the group developing (Dutra et al., 2018) a magnetic wine cooler, following the project “PoloMag”. As a matter of fact, during the Conference Thermag VIII, Dutra et al. announced (Dutra et al., 2018) to be working on an experimental test apparatus matching the working conditions of the HTF that would be provided to the heat exchangers by a hypothetical AMR. The device was connected to a cabinet (wine storing purposes) through a heat exchanger, with the purpose of investigating the energy performances for different temperatures of the secondary fluid coming from the AMR. Dutra et al. in their work (Dutra et al., 2018) did not publish any experimental data yet.

In 2013 Tusek et al., with the research group of University of Ljubljana, introduced (Tušek et al., 2013) a novel reciprocating magnetic refrigerator, visible in Figure 10, in which a magnet circuit of permanent magnets moves over a stationary AMR ($\Delta B = 1.15$ T); it reaches a maximum operating frequency of 1 Hz. Tušek et al. tested (Tušek et al., 2013) it while it was mounting 6 different configurations of Active magnetic Regenerators, occupied by gadolinium and crossed by 67% distilled water–33% ethylene glycol. During subsequent experiments Tušek et al. tested (Tušek et al., 2014) the prototype while it was working with $\text{LaFe}_{1-x-y}\text{Co}_x\text{Si}_y$ in three compositions and they observed $\Delta T_{\text{span,max}} = 20$ K under zero-load and 42 W as cooling power in correspondence of $\Delta T_{\text{span}} = 1$ K. This device is also well-described by Kitanovski et al. (Kitanovski et al., 2015c)

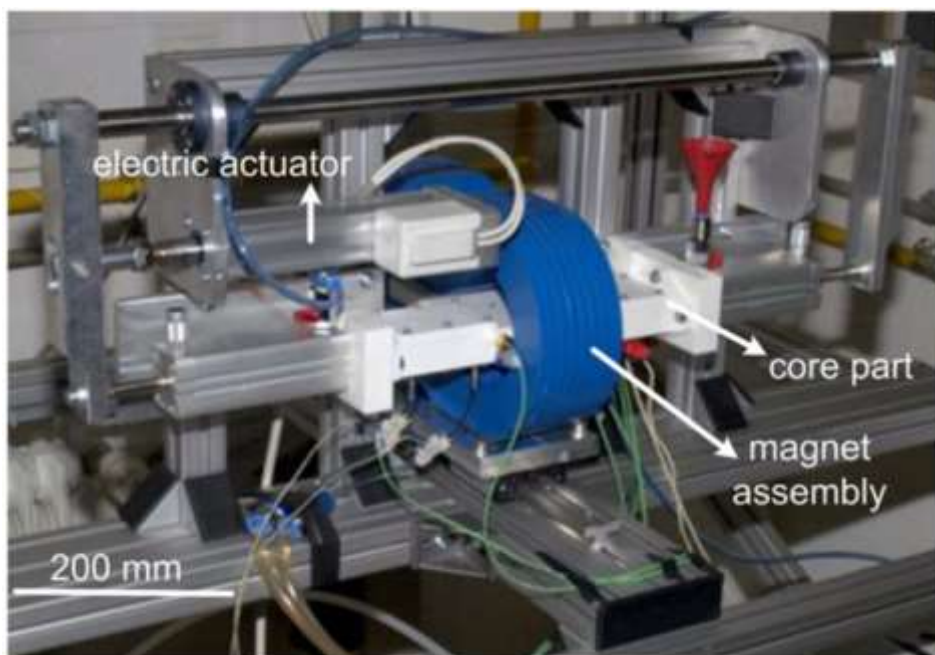


Figure 10. The Slovenian reciprocating prototype, built at University of Ljubljana (Tušek et al, 2013).

Recent updates about the University of Ljubljana research group were given in 2018 during Thermag VIII, with the announcement by Tomc et al. (Tomc et al., 2018) of being developing an electromagnets-based AMR wine-cooler ($\Delta B=1$ T). The device mounting AMR filled with Gd spheres (100-300 μm) and crossed by 70% distilled water–30% glycol mixture, can operate up to a frequency of 25 Hz. Tomc et al. tested (Tomc et al., 2018) it under no load conditions and saw it exhibiting $\Delta T_{\text{span,max}}=13$ K with $f=2$ Hz and $T_H=22\text{--}25$ °C. In 2012, during Thermag V, the American research group from Oak Ridge National Lab. presented (Shassere et al., 2012) initial experimental tests on a reciprocating AMR prototype mounting electromagnets ($\Delta B=0.75$ T) and operating on two regenerators made with gadolinium sheets which were crossed by a commercial water/glycol mixture. They showed (Shassere et al., 2012) that the device can produce 8.8 W kg^{-1} under $\Delta T_{\text{span}}=5.6$ K. About this prototype, the group did not provide further updates.

After having developed two reciprocating devices before 2010 (Yu et al., 2010), as reported by Kitanovski et al. (Kitanovski et al., 2015c), the Japanese Kobe University built in 2011 the first rotary device made of a permanent magnet magnetic system ($\Delta B=0.6$ T), mounting regenerator of packed bed particles of gadolinium that, working at 1 Hz as maximum AMR frequency, was able to produce $\Delta T_{\text{span,max}}=5$ K and $Q_{\text{ref,max}}=10$ W. Additionally, Kitanovski et al. listed (Kitanovski et al., 2015c) that the Japanese group, together with the Railway Technical Research Institute, developed in 2011 also another rotary device consisting of a magnet assembly in Halbach array configuration and that could achieve $\Delta T_{\text{span,max}}=7.6$ K, while the maximum cooling power was 150 W. Subsequently, in 2014, the group introduced (Hirano et al., 2014) a Next Generation Heat Pump System: a magnetic heat pump designed and produced using a cascading connection with multiple magnetocaloric materials, so that a maximum refrigeration capacity belonging to kW class was achieved. Hirano et al. said (Hirano et al., 2014) that a new design of Halbach array magnetic circuit permitted to reach $B_{\text{max}}=1.5$ T and that the device was able to work up to $\dot{V}=25\text{ l.min}^{-1}$; they registered 1.4

kW as highest cooling power under zero-temperature span and $\Delta T_{\text{span,max}}=21$ K under zero-load conditions, while the device was working with gadolinium alloys. Another remarkable Japanese device was built, in 2013, at the Sanden Co. Inc.; it composed of a magnetic system made of permanent-magnets ($\Delta B=0.85$ T) and it can achieve 13.8 K as maximum ΔT_{span} and a specific cooling power of 157.1 W kg^{-1} , as classified by Kitanowski et al. (Kitanowski et al., 2015c).

Arisen in 2009 with the announcement from University of Genoa of the first italian reciprocating device, already described by Yu et al. (Yu et al., 2010), the italian activities in magnetocaloric prototypes building continued in 2013 with the introduction by Tagliafico et al. (Tagliafico et al., 2013) of the second reciprocating permanent magnets prototype ($\Delta B=1.55$ T) employing gadolinium in a parallel-plates regenerator, crossed by a 50%water–50%ethanol mixture. Tagliafico et al. observed (Tagliafico et al., 2013) that the device performed 5 K as temperature span under no-load conditions.

Aprea et al. in 2014 developed (Aprea et al., 2014) 8-Mag, the first italian rotary device, shown in Figure 11: in the device a magnetic system made of permanent magnets shaped in double U design ($\Delta B_{\text{max}}=1.25$ T) moves circumferentially on a stationary ring on which 8 packed-bed regenerators are fixed (gadolinium spheres 0.4-0.5 mm diameter) and crossed by distilled water as HTF. Parallely to the experimental investigations on the prototype, some numerical tools were employed to study the performances and to indicate the way to go in the optimization of the prototype (Aprea et al., 2013; 2015a; 2018b). Aprea et al. observed (Aprea et al., 2014) the device exhibiting $\Delta T_{\text{span,max}}=13.5$ K under zero-load conditions, with $\Phi=1.40$ and $T_H=298$ K. Aprea et al., after further investigations on the device, observed (Aprea et al., 2016a) 8-Mag producing $\text{COP}=1.8$ in correspondence of a cooling power of 163 W ($f=0.5$ Hz, $\dot{V}=7.0 \text{ l min}^{-1}$ and $T_H=295$ K) and $\text{COP}_{\text{max}}=2.5$ under a cooling load of 200 W ($f=0.38$ Hz, $\dot{V}=5.0 \text{ l min}^{-1}$ and $T_H=296$ K) (Aprea et al., 2016b). Moreover, in 2015 Aprea et al. tested (Aprea et al., 2015b) 8Mag in air-conditioning operation mode, since they connected the hot heat exchanger to a geothermal probe, and they observed the device producing $\dot{Q}_{\text{ref}}=100$ W in correspondence of $\Delta T_{\text{span}}=6$ K and $\text{COP}=0.6$. In 2017, Aprea et al. introduced (Aprea et al., 2017) an artificial-neural-network control technique based, to optimize the performances of 8Mag.



Figure 11. 8Mag, the first italian rotary magnetic refrigerator (Aprea et al., 2016b).

In 2018, another italian prototype was built, following “FRIMAG” project the Emilia-Romagna Region (Italy), which involves the research groups of University of Bologna and IMEM-CNR; its design was introduced in 2018 at Thermag VIII Conference (Rosati et al., 2018): it is a rotary device composed by four gadolinium parallel-plates regenerators ($t_{\text{ck}}=0.25$ mm) magnetized and demagnetized by the rotation of a permanent magnet assembly ($\Delta B_{\text{max}}=1.4$ T). No details about the energy performances were provided yet by the authors.

The interest of the Spanish researchers toward the construction of a magnetocaloric prototype was underlined in 2013 when University of Coruna introduced (Gómez et al., 2013b) its reciprocating permanent magnets magnetocaloric prototype ($\Delta B=1\text{T}$), consisting of two AMR regenerators filled with parallel-plates of gadolinium in which distilled water flows (Figure 12). Gómez et al. declare (Gómez et al., 2013b) that the maximum temperature span measured was 3.5 K under zero-load condition and the specific cooling power at zero-temperature span was 16.7 W kg^{-1} . One year later, another Spanish University built a magnetocaloric device: during Thermag VI, researchers from University of Zaragoza announced (Velázquez et al., 2014) the realization of their magnetic refrigeration demonstrator, displayed in Figure 13: it is reciprocating and equipped with the magnetic system made of permanent magnets of $\text{Nd}_2\text{Fe}_{14}\text{B}$ in Halbach configuration

($\Delta B=1.3\text{--}1.4\text{ T}$) magnetizing a moving packed-bed regenerator filled with gadolinium spheres (0.2–0.4 mm) and crossed by a mixture of 75%water – 25%ethylene glycol as HTF. Velázquez et al. detected (Velázquez et al., 2014) a maximum temperature span of 19.3 K during no-load tests (with $\Phi=0.19$ and $\dot{m}=2.61\text{ g s}^{-1}$) and $\dot{Q}_{ref,max}=6\text{ W}$ during zero temperature-span investigations, with $\Phi = 1.1$ and $\dot{m} = 2.61\text{ g.s}^{-1}$.

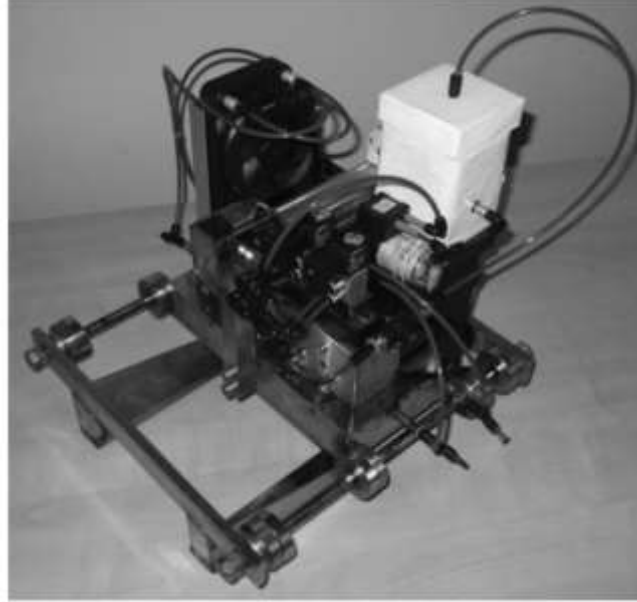


Figure 12. The Spanish magnetocaloric prototype developed by University of Coruna (Gómez et al., 2013b).

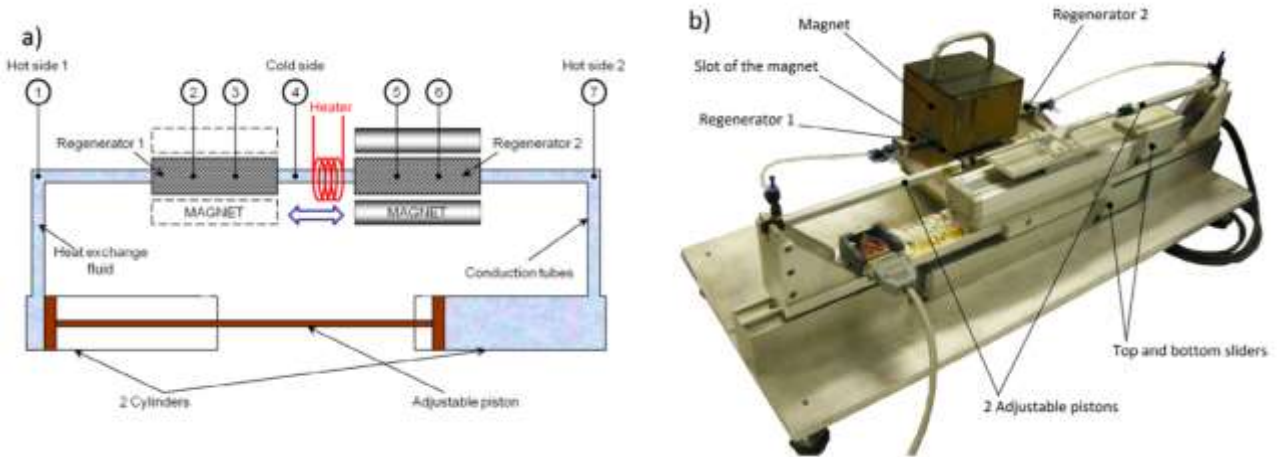


Figure 13. The Spanish magnetocaloric demonstrator developed by University of Zaragoza: (a) a conceptual scheme and (b) a picture of the complete system (Velázquez et al., 2016).

Jacobs et al., 2014 designed (Jacobs et al., 2014) for Astronautics Corporation of America a rotary magnetic refrigeration system, which is reported in Figure 14: it is based on a permanent magnets magnetic system ($\Delta B=1.44\text{ T}$) moving over 21 static regenerators assembled with LaFeSiH alloys and crossed by water. Jacobs et al. declared (Jacobs et al., 2014) the magnetic prototype can achieve $\dot{Q}_{ref,max}=3042\text{ W}$ during zero

temperature-span investigations and a promising cooling power of 2502 W in correspondence of $\Delta T_{\text{span}}=11$ K.



Figure 14. The magnetic refrigeration system built by Astronautics Corporation of America (Jacobs et al., 2014).

In the same year (2014), the research group of Wroclaw University of Technology introduced (Czernuszewicz et al., 2014a) a reciprocating magnetocaloric test stand, shown in Figure 15: the device mounts a Halbach array of permanent magnets ($\Delta B=1$ T) magnetizing an AMR packed-bed regenerator made of gadolinium within an ethane-diol mixture flows. Czernuszewicz et al. tested (Czernuszewicz et al., 2019) the device mounting six different regenerator geometries and they declared that the largest ΔT_{span} achieved is 6.9 K.

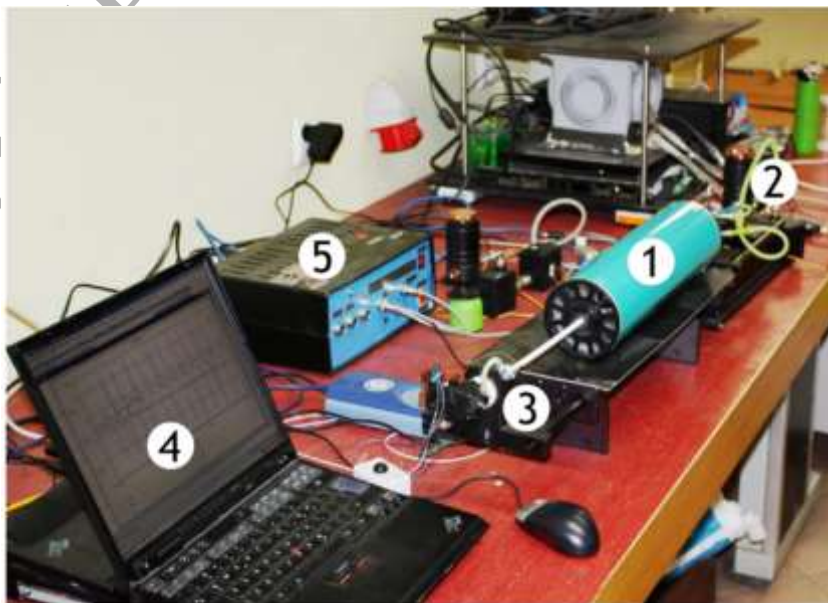


Figure 15. The magnetocaloric prototype built by Wroclaw University of Technology (Czernuszewicz et al., 2014a).

In 2016, during the Thermag VII, TU Delft in collaboration with South China University of Technology announced (Huang et al., 2016) to be working on the development of a rotary-type magnetic heat-pump setup: the device was assembled with a magnetic circuit ($\Delta B_{\text{mean}}=0.875$ T) rotating on 7 regenerators filled with packed bed of gadolinium spheres (0.4-0.8 mm) mounted on a fixed ring. Huang et al., during Thermag VIII, divulged (Huang et al., 2018) that the maximum achievable rotary frequency is 1.5 Hz and that, basing on preliminary tests with only two regenerators filled, they detected $\dot{Q}_{\text{ref,max}}=40$ W in correspondence of 9.1 K as temperature span. A picture of the prototype is reported in Figure 16.

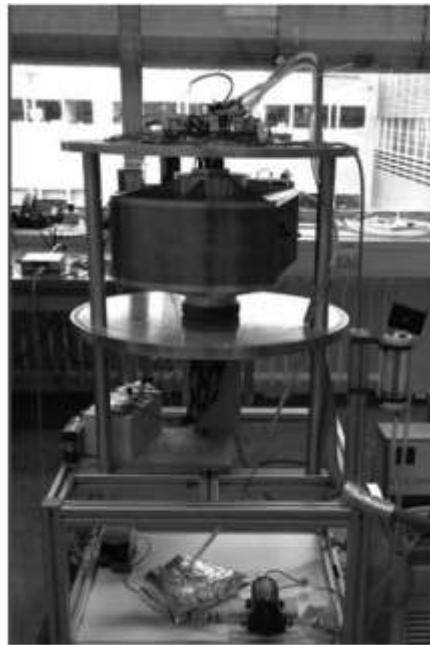


Figure 16. The rotary prototype developed by TU Delft in collaboration with South China University of Technology (Huang et al., 2018).

During Thermag VII, not only Nederland introduced its first magnetocaloric prototype; Sweden did it too, with KTH Royal Institute of Technology: Monfared and Palm introduced (Monfared and Palm, 2016) a device with rotating magnet system (1.1 T) on 12 static irregular-shape packed bed AMR made of La-Fe-Mn-Si-H. Monfared and Palm investigated (Monfared and Palm, 2016) the device with only 2 of 12 regenerators filled and they observed a temperature span of around 18 K, for $\dot{V}=8.3 \times 10^{-6} \text{ m}^3 \text{ s}^{-1}$ and $f=0.5$ Hz. Another prototype presented at Thermag VII was developed from the Chinese Baotou Research Institute of Rare-earths (Zhang et al., 2016) and it was a magnetic cabinet (wine storage purposes) employing two concentric Halbach array permanent magnets ($B_{\text{max}} = 1.5$ T) rotating over a multi-layer AMR made of Gd and GdEr alloy particles whereas demineralized water was used as HTF. Zhang et al. found (Zhang et al., 2016) $\Delta T_{\text{span,max}}=25$ K at zero-load while operating at the optimal frequency of 2 Hz and $\dot{V}=5.3 \text{ l min}^{-1}$.

During the following Thermag VIII, in 2018, the German research group of TU Darmstadt exhibited (Benke et al., 2018) its green magnetic cooling device (shown in Figure 17): the first magnetocaloric demonstrator based on recycled permanent magnets in order to improve the ecological footprint of magnetocaloric technology. Benke et al. said (Benke et al., 2018) that the employed materials were alloys of La-Fe-Si and that by stacking 5 different compounds it was possible to increase the achievable thermal span from 8.1 K to 20.6 K, even if they did not provide any experimental results yet.

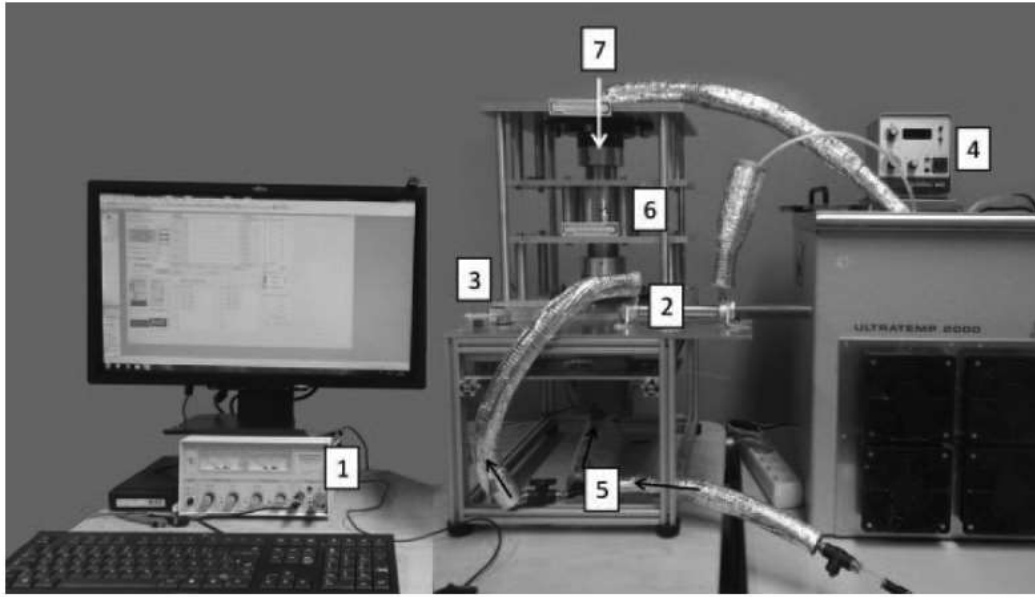


Figure 17. The green magnetic cooling device presented by TU Darmstadt research group (Benke et al., 2018).

In the same conference, also Turkey ratified its entry among the magnetocaloric prototype developers, introducing two devices (Baser et al., 2018a; 2018b), visible in Figure 18. The first is a reciprocating magnetic refrigerator called MAGICE-1 (Baser et al., 2018a) that includes one cylindrical shaped AMR, filled with Gd spheres (0.5 mm) through which a mixture of 80%water-20%ethylene glycol flows, moving alternatively under a magnetic system realized with permanent magnets in Halbach array configuration. Baser et al. announced (Baser et al., 2018a) that the maximum no-load ΔT_{span} was 12 K, whereas $\text{COP}_{\text{max}}=2$ was measured in correspondence of $\dot{Q}_{\text{ref}}=75$ W with $f=0.1$ Hz. The second device is a rotary magnetic refrigerator that consists (Baser et al., 2018b) of 24 packed regenerators subjected to the rotation of a four-pole permanent magnets magnetic system. Baser et al. performed (Baser et al., 2018b) preliminary tests in a wide set of conditions ($f=0.1\div 2$ Hz, $T_{\text{H}}=290\div 310$ K and $\dot{V}1.5\div 6$ l min⁻¹) and they detected no cooling-load $\Delta T_{\text{span,max}}=5$ K and $\text{COP}_{\text{max}}=3.5$ at $\dot{Q}_{\text{ref}}=500$ W (with $f=0.1$ Hz, $\dot{V}=2.8$ l min⁻¹ and $T_{\text{H}}=300$ K).

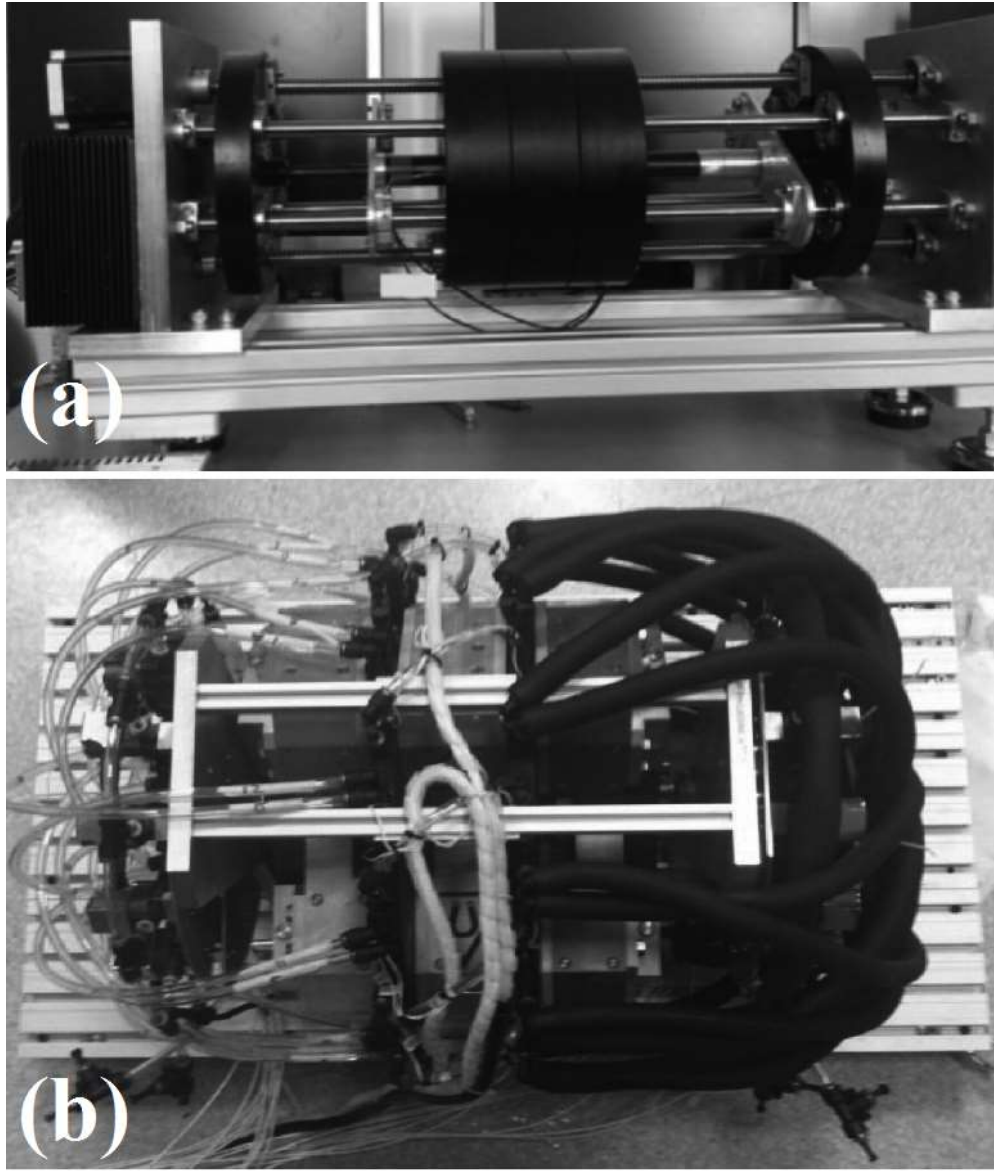


Figure 18. The the Turkish (a) reciprocating and (b) rotary prototypes (Benke et al., 2018a; 2018b).

Next to the interest from the universities and academic research groups toward magnetocaloric cooling and heat pumps, there are also some companies well-active in this field. One of these is the Cooltech Application Company that patented in 2009 its first prototype, a reciprocating AMR, already described and listed by Yu et al. (Yu et al., 2010). As reported by Kitanovski et al. (Kitanovski et al., 2015c), in 2011 this company developed their second reciprocating device, made with Nd-Fe-B permanent magnets ($\Delta B=1.27$ T) able to operate up to 1.5 Hz: it mounts a parallel-plate AMR assembled with Gd and Gd-Tb and achieving $\dot{Q}_{ref,max}=150$ W and $\Delta T_{span,max}=38$ K. Moreover, the overview of Kitanovski et al. (Kitanovski et al., 2015c) lists that the Cooltech Application Company developed: in 2013 its first rotary permanent magnets magnetic prototype ($\Delta B=0.98$ T) up to reach 4 Hz by the rotation on packed bed regenerators (Gd or Gd-Tb) and producing $\Delta T_{span,max}=42$ K and $\dot{Q}_{ref,max}=120$ W; in 2014, the second Cooltech rotary prototype ($\Delta B=1.17$ T), mounting parallel-plates regenerators (tested with Gd, Gd-Er/La-Fe-Si) and touching $\Delta T_{span,max}=38$ K and $\dot{Q}_{ref,max}=300$ W. The latest updates about the Cooltech Application Company were presented by Chaudron

et al. (Chaudron et al., 2018) during Thermag VIII, with the introduction of a rotary device, reported in Figure 19 and conceived for a possible replacement of Gd-based with LaFeSi-based alloys. Chaudron et al. declare (Chaudron et al., 2018) that the prototype has three thermal loops and a hydraulic system moved by an external pump; the magnetic system is made of permanent magnets ($\Delta B=1$ T) and the device can operate between $f=0.5\div 2$ Hz and variable flow rates $1\div 6$ l min⁻¹. The Company states (Lionte et al., 2018b) that during initial investigations the prototype produces $\Delta T_{\text{span,max}}=30.8$ K.



Figure 19. The proof-of-concept of Cooltech with the integration of a cabinet (Chaudron et al., 2018).

Table 2 summarizes schematically all the magnetocaloric devices built from the year 2010 to 2019, highlighting and reporting the salient information from both a design and a performance point of view.

TABLE 2

4. Classification of the electrocaloric devices operating at room temperature

The electrocaloric effect was observed for the first time by Kobeko and Kurtschatov in 1930 in Rochelle salt (Kobeko & Kurtschatov, 1930; Correia & Zhang, 2014; Moya et al., 2014). In 1943 ECE was first-time quantitatively measured by Hautzenlaub (Correia & Zhang, 2014). These works triggered the interest toward electrocaloric field and huge efforts were made in developing new materials that could be promising for cooling, mainly devoted to cryogenic applications (Lawless, 1977; Lawless and Morrow, 1977). In 1979 the first concept of cryogenic heat-switches based electrocaloric refrigerator was developed and tested by Radebaugh et al. (Radebaugh et al., 1979). It employed SrTiO₃ ceramics showing an electrocaloric

temperature change of 0.3 K under $\Delta E = 20 \text{ kV cm}^{-1}$ and operating in a $T_C - T_H = 4\text{--}15 \text{ K}$ range. The concept of electrocaloric refrigerators employing Active Electrocaloric Regenerator was introduced primarily in 1989 by Moscow Power Engineering Institute (URSS) with its electrocaloric refrigerator prototype, shown in Figure 20, mounting multilayer of $\text{PbSc}_{0.5}\text{Ta}_{0.5}\text{O}_3$ (PST) ceramic with phase transition temperatures shifted in successive decreasing up to 250 K (Sinyavsky et al., 1989). Sinyavsky & Brodyansky conducted (Sinyavsky & Brodyansky, 1992) investigations in room-temperature range on the device mounting a PST regenerator crossed by liquid pentane as HTF and subjected to $\Delta E = 60 \text{ kV cm}^{-1}$. Sinyavsky & Brodyansky concluded (Sinyavsky & Brodyansky, 1992) that a maximum temperature span of 5 K was achieved under $\Delta E = 60 \text{ kV cm}^{-1}$ with $T_H = 300 \text{ K}$ and $f = 5 \text{ Hz}$.

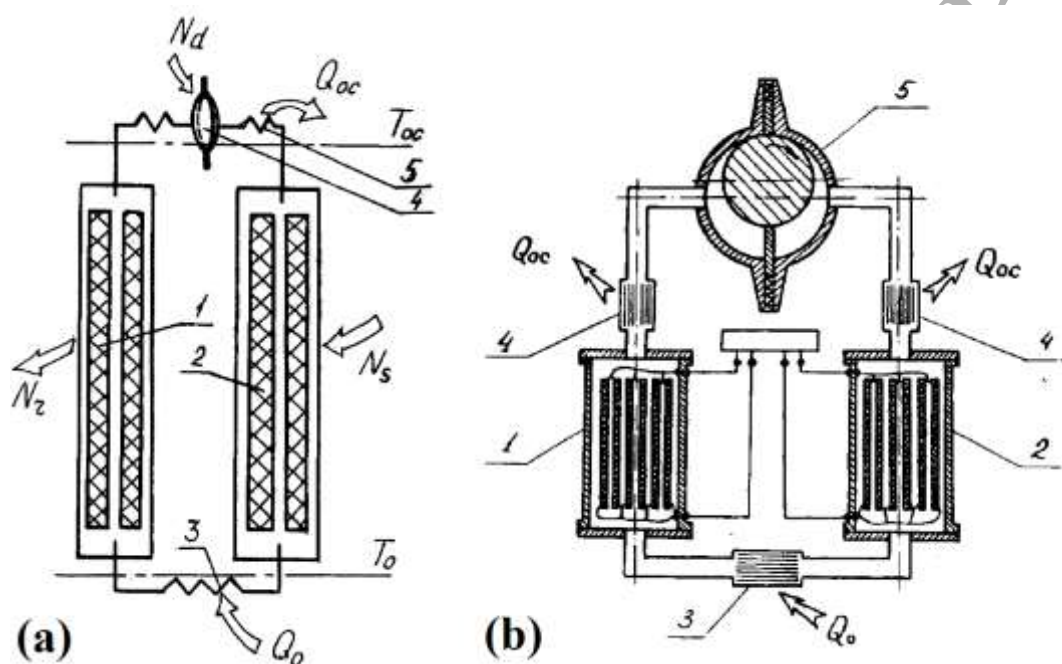


Figure 20. The electrocaloric refrigerator concept developed by the Moscow Power Engineering Institute: (a) schematic of the AER (Sinyavsky et al., 1989); (b) schematic of the electrocaloric device (Sinyavsky & Brodyansky, 1992).

In 2005, Lawless patented (Lawless, 2005) a thermal transfer device based on the electrocaloric effect exhibited by $x(\text{PbMg}_{0.33}\text{Nb}_{0.67}\text{O}_3) + y(\text{PbSc}_{0.5}\text{Nb}_{0.5}\text{O}_3)$ or by $x(\text{PbMg}_{0.33}\text{Nb}_{0.67}\text{O}_3) + y(\text{PbTiO}_3) + z(\text{SrTiO}_3)$. The device was able to work in a temperature range $[-10; +50]^\circ\text{C}$ and it was projected to work in single or multilayer capacitor configuration including electrodes and the electrocaloric material as the capacitor dielectric.

In the year 2014, the research group of University of Ljubljana, during Thermag VI, presented (Plaznik et al., 2014) their small scale electrocaloric cooling device with an AER in which lead magnesium niobate-lead titanate (PMN-PT) ceramic plates were used as ECE material of the regenerator. In the device, whose details are displayed in Figure 21, is assembled a parallel-plate AER made of $30 \text{ } 0.90(\text{PbMg}_{1/3}\text{Nb}_{2/3}\text{O}_3) - 0.10(\text{PbTiO}_3)$ bulk ceramic slices (PMN-10PT) ($0.2 \text{ mm} \times 10 \text{ mm} \times 20 \text{ mm}$) through which a dielectric fluid (silicone oil) vehiculates heat fluxes by a pump. Plaznik et al. performed (Plaznik et al., 2014) preliminary

tests in which the AER was subjected to $\Delta E = 2.5 \text{ MVm}^{-1}$ and they observed a $\Delta T_{\text{span,max}} = 2.1 \text{ K}$ with $f = 1.25 \text{ Hz}$. After further experiments, in 2015, Plaznik et al. registered (Plaznik et al., 2015) maximum temperature spans of 2.3K and 3.3K under 2.5 and 5 MV m^{-1} , respectively, with the device working at the operating frequency of 0.75 Hz. In 2019, the group introduced (Plaznik et al., 2019) an upgraded version of the device, in which AER was formed by five stacks of PMN-10PT and each stack was constructed from 9 EC plates positioned on top of one another: with $f = 0.65 \text{ Hz}$, $\Delta E = 57 \text{ kV cm}^{-1}$ and $T_H = 296 \text{ K}$, $\Delta T_{\text{span,max}} = 3.3 \text{ K}$ for $V^* = 0.2$, and a maximum specific cooling power of 16 W kg^{-1} in correspondence of $\Delta T_{\text{span}} = 1 \text{ K}$, were noticed.

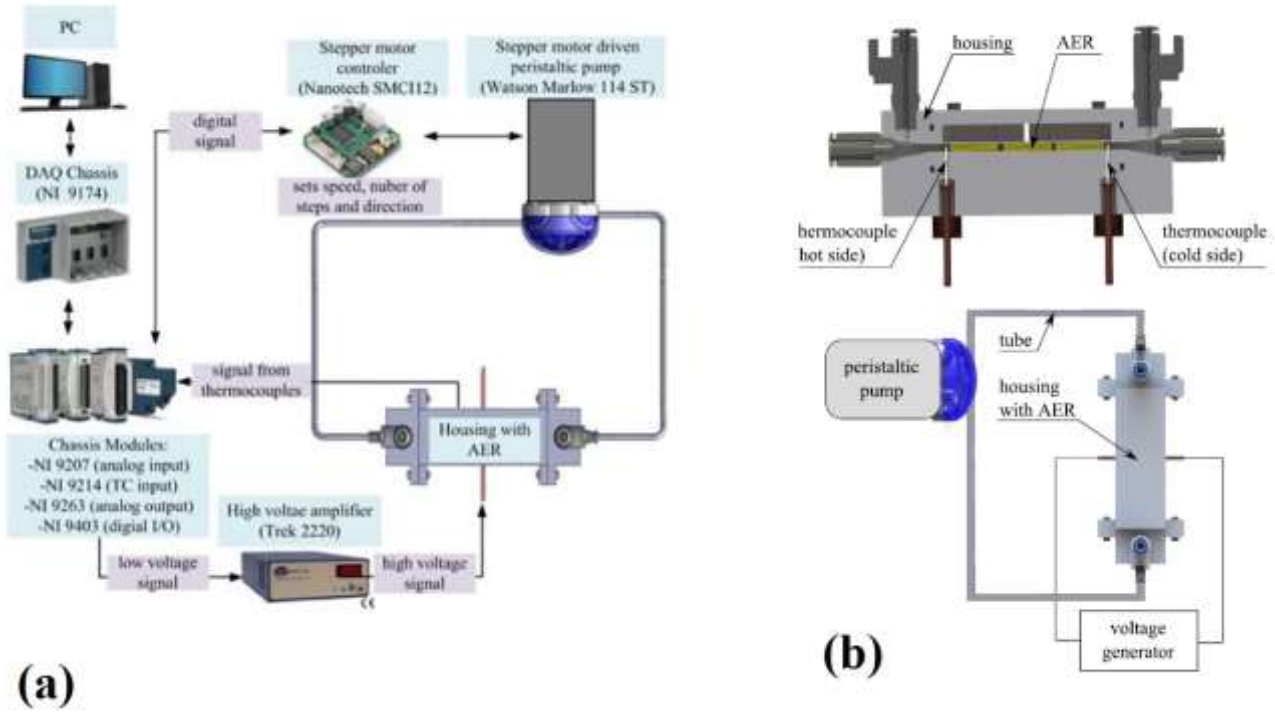


Figure 21. The small-scale electrocaloric cooling device with developed by University of Ljubljana: schematic presentation of the (a) experimental loop for AER performance testing; (b) small-scale cooling (Plaznik et al., 2014; 2015).

In 2016, Blumenthal et al. presented (Blumenthal et al., 2016) to scientific community an ECE cooling device assembling an AER built with relaxor ferroelectric elements, magnesium niobate-lead titanate (PMN-PT). The prototype, defined as demonstrator, was conceived for battery systems refrigeration purposes; indeed, the components are designed for operation around 313K. The device, shown in Figure 22, consists of an AER placed between two heat exchangers; the heat transfer was regulated by silicon oil moved by pistons pushing it through the AER channels ($t_{\text{ck}} = 0.8 \text{ mm}$). Blumenthal et al. attested that (Blumenthal et al., 2016) the device could operate in the frequency range $0.1 \div 1.4 \text{ Hz}$ and that the maximum ΔT_{span} detected was 0.17 K at $\Delta E = 1 \text{ kVmm}^{-1}$, $f = 1 \text{ Hz}$.

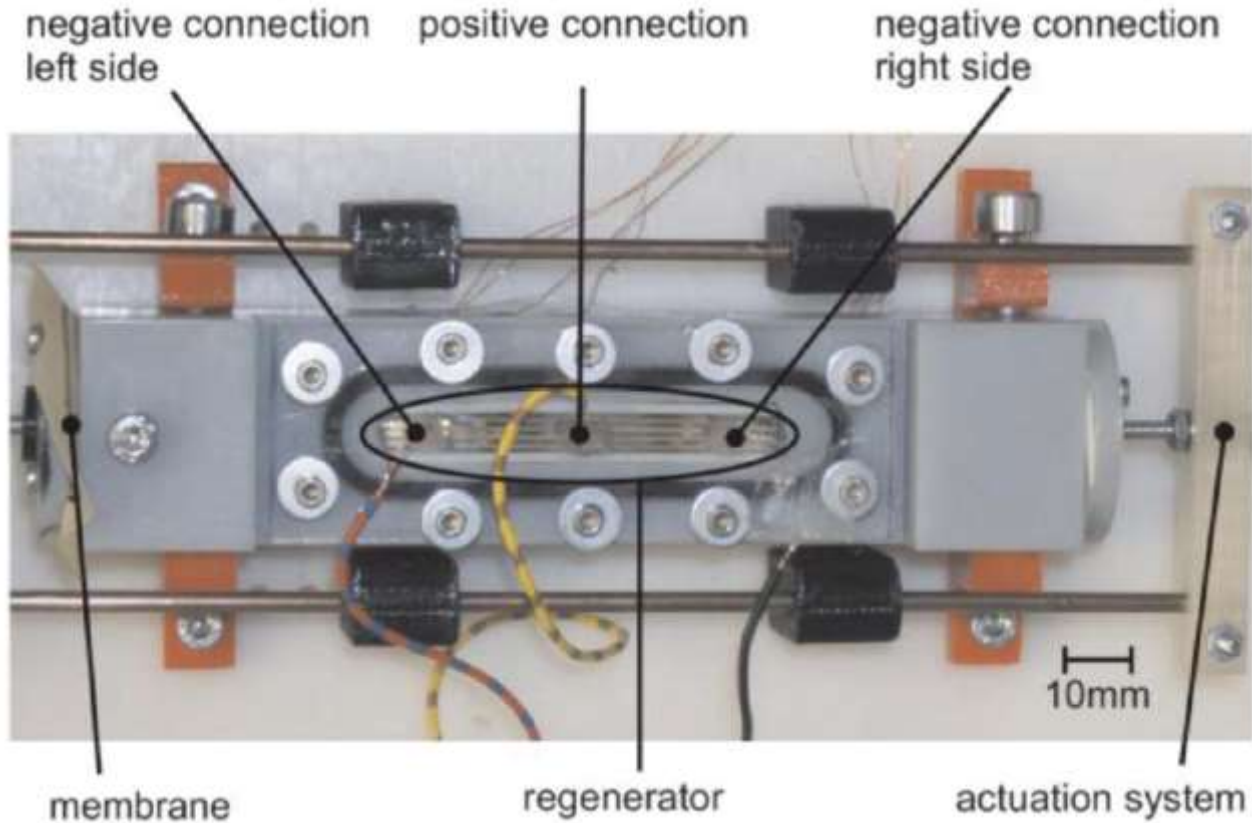


Figure 22. The AER device demonstrator built by Leibniz Universität Hannover (Blumenthal et al, 2016).

Parallely, a small number of solid-to-solid thermal diode-based electrocaloric prototypes were developed in a very recent past; since the attention in this paper is focused mostly on solid-state caloric effect active prototypes, they will not be described in detail but, for completeness, only mentioned. The Pennsylvania State University developed two thermal diode-based devices: in 2013 Gu et al. realized (Gu et al., 2013) a solid-to-solid regenerator achieving a temperature span higher than 6 K under $\Delta E = 100 \text{ MV m}^{-1}$. In 2017, Zhang et al. reported (Zhang et al., 2017) on a prototype consisting of two rings of ECE grounded on commercial Y5V ceramic, able to reach $\Delta T_{\text{span,max}} = 2 \text{ K}$ under $\Delta E = 16.6 \text{ MV m}^{-1}$. During Thermag 2018, Parc, a Xerox company presented (Wang et al., 2018) a self-regenerating electrocaloric cooler, made of two layers of alternating rows of ECE and insulating material and achieving, in preliminary results, $\Delta T_{\text{span,max}} = 2.5 \text{ K}$ under $\Delta E = 30.8 \text{ MV m}^{-1}$.

A resume of the electrocaloric devices built could be found in Table 3.

TABLE 3

5. Classification of the elastocaloric devices operating at room temperature

Elastocaloric effect is used for referring to the reversible thermal response detected in an elastocaloric solid-state material because of an uniaxial stress-induced elongation. The interest toward such technique for cooling or heat-pumping applications is very recent: from the beginning of the new millennium a

conspicuous number of researches devoted to the investigation of the physical properties of the elastocaloric materials was collected (Qian et al., 2016a).

As attested by Kirsch et al. in their review on elastocaloric cooling (Kirsch et al., 2018), the very first elastocaloric cooling demonstrator was presented (Cui et al., 2012) at the ARPA-E Energy Innovation Summit Technology Showcase in 2012 by University of Maryland in collaboration with Pacific Northwest National Lab. In this device NiTi wires (0.5 mm diameter), crossed by air as HTF, are installed on two annular rings, whose synchronous rotation determines the loading-unloading processes due to the inclination angle of the upper ring. Cui et al. observed (Cui et al., 2012) the device achieving $\Delta T_{\text{span,max}}=17$ K under 8.5% for stretching, 6.0% for compression.

The year 2015 was characterized by an increasing number of realized elastocaloric devices. Schmidt et al. realized (Schmidt et al., 2015) an elastocaloric proof-of-concept for solid-to-solid eCE refrigeration purposes. The device, shown in Figure 23, is set up in two levels: in the upper one there is the linear direct driver to load and unload the material and a load cell is assembled on the edge of the prototype to measure the applied mechanical stress; the lower level realizes the heat transfer between the eCE-material and the heat exchangers.

A second linear driver switches the positions of the hot and the cold heat exchangers while the material is moving orthogonally to the first linear driver. During experimental investigations with the device employing Ni-Ti alloys, Schmidt et al. observed (Schmidt et al., 2015) a maximum $\Delta T_{\text{span}}=7$ K at a contact time of 1.2 s with 4% strain.



Figure 23. The first elastocaloric test setup device for elastocaloric cooling (Schmidt et al., 2015).

In 2015, University of Maryland, in collaboration with Niron Magnetics Inc., also introduced (Qian et al., 2015a) their device to scientific community, announcing its development during 4th IIR International Congress of Refrigeration. The prototype, based on a novel heat recovery mechanism, consisted of two beds filled with nitinol tubes, crossed by water and subjected to compression-decompression cycles through a linear actuator. Qian et al. declared (Qian et al., 2015a) that initial results produced a maximum $\Delta T_{\text{span}}=1.5$ K and 38 W of cooling power. Following, a novel design was introduced (Qian et al., 2015b) with the purpose

of reducing the cyclic losses caused mostly by unnecessary dead-thermal-masses inside the system and a multi-objective optimization was applied by a genetic algorithm to explore the potential of such technique. Qian et al. registered (Qian et al., 2016b) $\Delta T_{\text{span,max}}=4.7$ K and $\dot{Q}_{\text{ref,max}}=65$ W. Subsequently, a further optimization of the prototype design was introduced with a single-stage hydraulically driven prototype, shown in Figure 24. The Ni-Ti tubes were limited by the holder and compressed axially by two sets of loading head assemblies, which were used to make the HTF flowing homogeneously into each tube of NiTi and to transfer the high compressive force. Qian et al. still did not introduce (Qian et al., 2016b) any experimental results directly carried out, but, using a numerical model, they found the maximum achievable $\Delta T_{\text{span}}=24.6$ K and $\text{COP}_{\text{max}}=11.0$ under 4.5 % strain.



Figure 24. The elastocaloric regenerator designed by University of Maryland (Qian et al., 2016b).

In 2016, Technical University of Denmark in collaboration with University of Ljubljana presented (Tušek et al., 2016) a regenerative elastocaloric device mounting a regenerator, shown in Figure 25(a), with thin “dog-bone-shaped” Ni-Ti plates loaded in tension with the maximum applied strain limited to 4% and therefore not corresponding to a fully complete transformation, to increase fatigue life of the device. The regenerator is crossed by HTF (water), unidirectionally. Tušek et al. tested (Tušek et al., 2016) the device under heat-pump operation mode and they observed $\Delta T_{\text{span,max}}=15.3$ K in correspondence of $\frac{\dot{Q}_{\text{ref}}}{m}=782$ W kg⁻¹ and COP=3.5 (with $f=0.25$ Hz and applied strain of 3.4%). Further updates of the prototype, presented by Engelbrecht et al. (Engelbrecht et al., 2017), see the system modified to provide more flexible control of the fluid flow and to reduce the total mass of HTF; the introduction of a novel regenerator, visible in Figure 25(b), constituted by 10 Ni-Ti plates ($t_{\text{ck}}=0.35$ mm); higher maximum achievable frequencies of operation. Engelbrecht et al. attested (Engelbrecht et al., 2017) that the system, displayed in Figure 26, reached a maximum $\Delta T_{\text{span}}=19.9$ K with an applied strain of 3.5% and $T_{\text{H}}=308$ K; they said (Engelbrecht et al., 2017) that this is the highest temperature span for an elastocaloric system. The strongest weakness point of elastocaloric systems is the fatigue-life of the elastocaloric material employed: the regenerators tested by Engelbrecht et al. (Engelbrecht et al., 2017), failed approximately before 6000 AeR cycles.

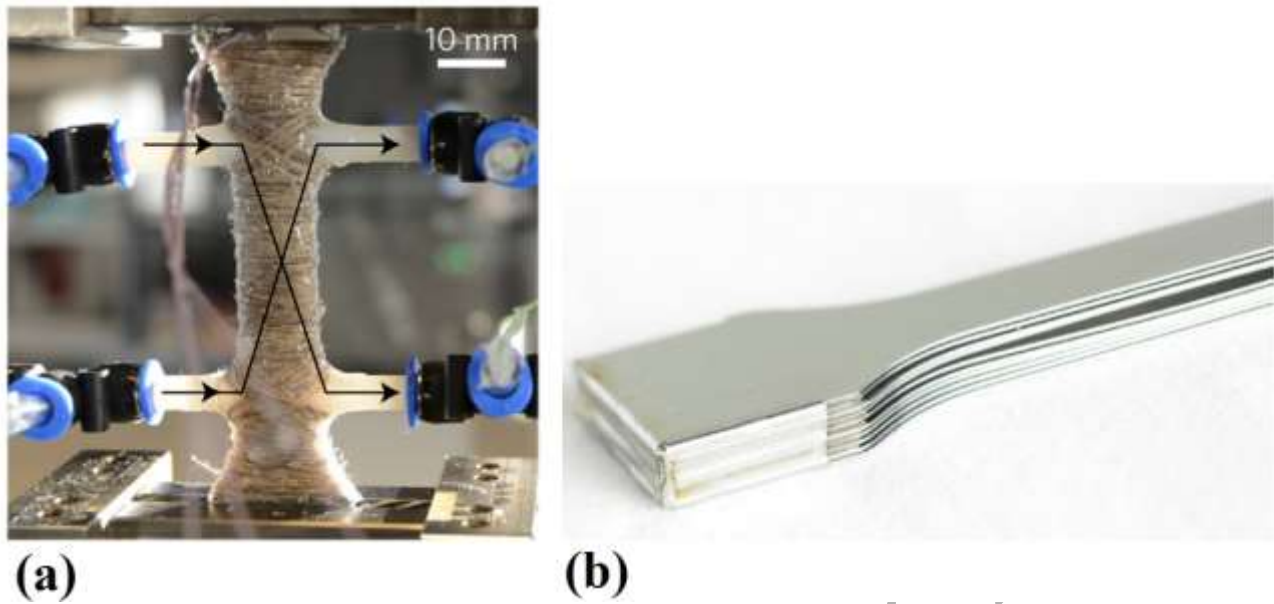


Figure 25. (a) First version of the regenerator, clamped in mechanical grips and connected to the fluid loops (Tušek et al., 2016); (b) the second design of the regenerator (Engelbrecht et al., 2017).

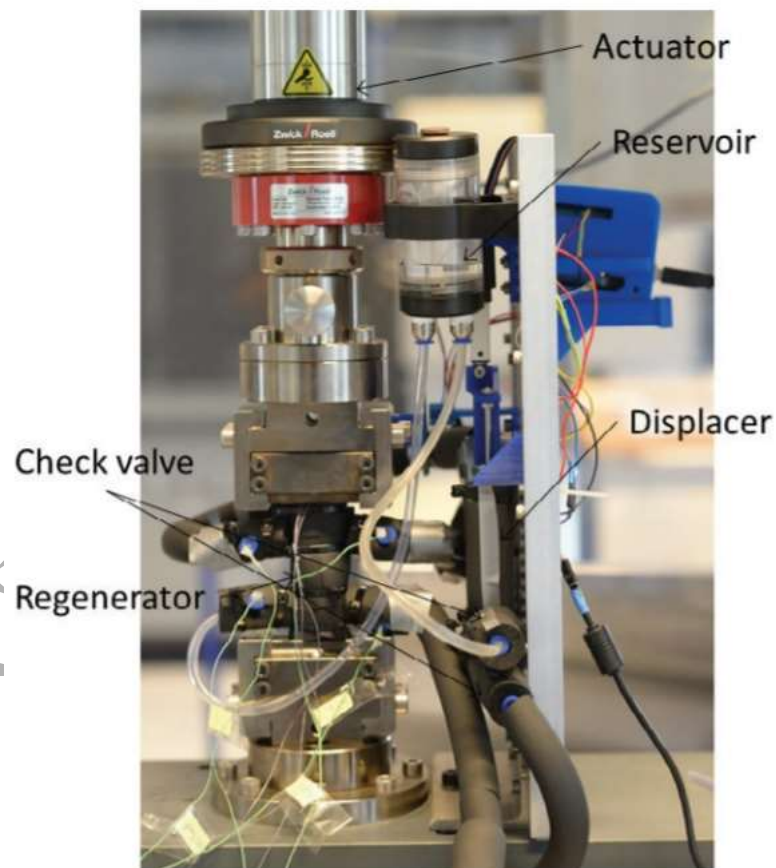


Figure 26. A picture of the AeR cooling device designed by Engelbrecht et al. (Engelbrecht et al., 2017).

In 2018 Kirsch et al. presented (Kirsch et al., 2018) a continuously operating NiTi-based elastocaloric cooling device to investigate the performance under real environmental conditions. The device is conceived as two co-rotating rings coupled with elastocaloric wires settled axially around them; one of the rings is

inclined to convert the rotary motion to a sinusoidal linear stroke. The eCE wires (NiTiCo-based alloys) were subjected to stress and unstress processes to produce a circular temperature distribution lengthwise the prototype. The disk made of elastocaloric materials moves in a duct consisting of an outer and inner cylinder, in which air flows as heat-transfer medium provoking the continuous extraction of cooling and heating powers. A scheme of the prototype design is visible in Figure 27.

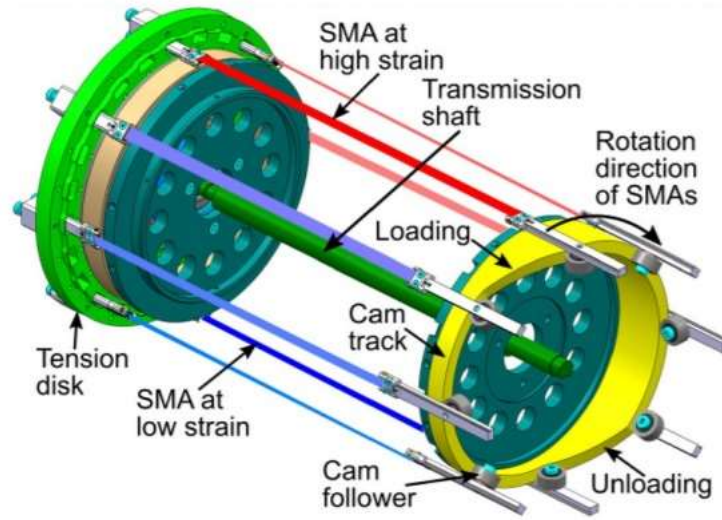


Figure 27. A schematic of the novel rotary elastocaloric cooling device. (Kirsch et al., 2018).

Based on initial experimental results conducted with 4.7% as maximum strain for loading-unloading processes, Kirsch et al. registered (Kirsch et al., 2018) a $COP=9.5$ in correspondence $\dot{Q}_{ref}=250$ W and $\Delta T_{span}=10$ K.

Up to now we have discussed only about the macroscale realizing of elastocaloric devices. Some research groups devoted their attention to the microscale for elastocaloric cooling. In 2015, Ossmer et al. introduced (Ossmer et al., 2015) their miniature elastocaloric solid-to-solid heat-pump employing $Ti_{50.5}Ni_{49.1}Fe_{0.4}$ foils ($t_{ck}=30\mu m$) mounted in a double bridge layout bonded to an alumina substrate, as visible in Figure 28. A spacer is attached to the center of one bridge and then the second bridge is pressed and bonded onto the first one. The heat exchangers of the prototype are assembled with three individual copper segments connected to a respective temperature sensor. During the operation, the elastocaloric material is periodically moved between the source and the sink surfaces; indeed, the heat transfer mechanism was realized by the contact between the solid pieces (without HTF) and, in 2015, Ossmer et al. registered (Ossmer et al., 2015) 7 K as maximum temperature span and 2.8 as COP_{max} . Further design optimizations (Ossmer et al., 2016) lead the device, in 2016, to achieve $\Delta T_{span,max}=9.4$ K; moreover, in 2017, Bruederlin et. al. detected (Bruederlin et al., 2017) $\Delta T_{span,max}=13$ K and they declared the device can achieve 10% of Carnot efficiency as maximum coefficient of performance.

In 2018, during Thermag VIII, the group presented (Bruederlin et al., 2018) the concept of cascaded microscale elastocaloric cooling system with a device mounting three film-based cooling units like the ones

presented before, coupled in a serial way to form a cascaded elastocaloric system, with a $\Delta T_{\text{span,max}}=14.8$ K touched at $f=2$ Hz.

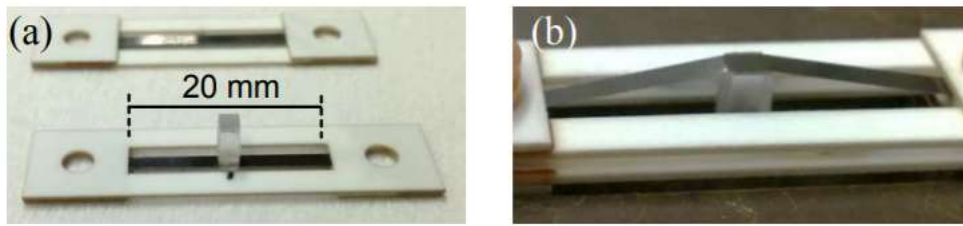


Figure 28. Integration technology for TiNi bridge structure. (a) TiNi bridges bonded on alumina substrates. (b) Double bridge (Ossmer et al., 2015).

Table 4 summarizes the history of elastocaloric prototypes-building.

TABLE 4

6. Classification of the barocaloric devices operating at room temperature

Barocaloric effect occurs as the thermal response to a hydrostatic pressure of a barocaloric material to an application. The barocaloric-focused researches aiming to the realization of devices for cooling and heat-pump applications are at the beginning, since barocaloric, among the caloric effect-based techniques, is still in an embryonic state. Literature did not contain experimental results coming out from the operation of barocaloric devices, yet. Up to now, we account only two ideas of prototype design for barocaloric applications. The first one was introduced by Czernuszewicz et al. (Czernuszewicz et al., 2014b) proposing the idea of a magneto-barocaloric test-stand for cooling applications, which should allow assessment of the application potential of the materials presenting a change of the temperature under the influence of magnetization and pressure application. No experimental results were provided.

The second and the latest idea of barocaloric devices was introduced in 2018 during Thermag VIII by University of Cambridge (Stern-Taulats et al., 2018) when they presented the design of the building challenges of the first barocaloric cooler that operates at room temperature, under small applied pressures. The system will operate with a barocaloric material made of cheap abundant elements manifesting a giant barocaloric effect. The challenge is to dispose of an efficient barocaloric device, lowering the values of the working pressure and optimizing the heat transfer processes. Stern-Taulats et al. ensured (Stern-Taulats et al., 2018) that the performance of the device will shortly be tested and we, as part of caloric scientific community, hope to appreciate the experimental results soon, in order to see caloric-cooling research taking another important step.

7. Conclusions

In this paper we gave an overview as wide as possible of the progresses made by solid-state caloric effect-based heating and cooling community in device realization. Retracing the whole history of prototypes development, beginning from magnetocaloric, we noticed how the research is characterized by sudden and

decisive steps that represent a breakthrough from the past to the future. To cite some examples, the introduction of rotative permanent magnets systems has drastically changed the conception of magnetocaloric device designing, or the discovery of Giant Electrocaloric Effect gave way to the interest in employing such phenomenon for cooling applications.

A lot of steps were taken particularly in magnetocaloric-prototypes development, since we have been able to appreciate the realization of devices that can produce cooling powers up to 3042 W (Jacobs et al., 2014), or temperature spans as high as around 40 K. The most recent elastocaloric refrigeration is making considerable progress too, reaching in the last five years of developing, the promising 20 K as highest temperature span, declared by Engelbrecht et al. (Engelbrecht et al., 2017). The most stringent bottleneck of elastocaloric is represented by is the fatigue-life of the elastocaloric materials employed; it currently does not allow the construction of long-lasting devices. Further efforts must be made in the field of the development of new eCE materials whose use for refrigeration and heat-pump purposes can overcome this obstacle. Less remarkable results were given in electrocaloric-device-developing field, based on active regeneration, because of the high losses related to electric-field generation that drastically bring down the energy performances of the realized prototypes. Barocaloric seems to be a promising and worthy of investigation technology on which caloric scientific community is now laying the first bases for the realization of heating and cooling devices.

Nowadays, the solid-state scientific community is moving toward further promising directions, alongside the world of active regeneration. One of these is represented by thermoelectric switches employed for magnetocaloric and electrocaloric coolers applications. Such technology has the advantage of presenting no moving parts that allows the miniaturizations and the operation of the devices at higher frequencies (Tomc et al., 2013). Xu et al. introduced an interesting perspective study on the synergic application of magnetocaloric and thermoelectric for cooling (Xu et al., 2018). A number of patents and investigations on thermoelectric switches for magnetocaloric cooling is available in open literature (Seeler & Degen, 2016; Ihnfeldt et al., 2017; Almanza et al., 2018; Goenka, 2018; Adams et al., 2019). With reference to ECE, a hopeful inherent work was presented by Feng et al. where a model of a small-scale heat pump based on thermoelectric and electrocaloric materials was introduced together with its energy performances (Feng et al., 2016). Another way is the simultaneous employment of several caloric effects for cooling, generated by different-nature fields and called multicaloric effect. Most of the studies published up to now are still theoretical; but more practical applications for cooling purposes are represented by Starkov and Starkov that modeled an efficient solid-state cooler based on layered multiferroics (Starkov & Starkov, 2014) and by Gottschall et al. (Gottschall et al., 2018) that proposed a proof of concept based on multicaloric cooling cycle in which also thermal hysteresis is accounted.

Downstream of all this, what we can say is that solid-state refrigeration and heat pumping technology has made great strides in the last decade, and we hope that it will soon be able to make the decisive one for carrying all the efforts done toward a large-scale implementation as well as the competitiveness of the solid-state prototypes based on caloric effects.

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Table 1. Overview of all the past Thermag Conferences held and the programmed next one.

Name	Location	Dates
THERMAG	Montreux, Switzerland	28-30 September 2005
THERMAG II	Portoroz, Slovenia	11-13 April 2007
THERMAG III	Des Moines, WA, United States	11-15 May 2009
THERMAG IV	Baotou, China	23-27 August 2010
THERMAG V	Grenoble, France	17-20 September 2012
THERMAG VI	Victoria, Canada	07-10 September 2014
THERMAG VII	Turin, Italy	11- 14 September 2016
THERMAG VIII	Darmstadt, Germany	16-20 September 2018
THERMAG IX	College Park, MD, United States	07-10 June 2020

Table 2. Magnetocaloric devices built to date and research group, and introduced after the year 2010, listed with their general characteristics.

Research group and location	Year Name	Type	$\dot{Q}_{ref,max}$ [W]	$\Delta T_{span,max}$ [K]	Magn. system ΔB_{max} [T]	MCM + HTF
University of Victoria, Canada (Tura and Rowe, 2011; Arnold et al., 2014)	2009 PM-I	Rotary	50	29	Permanent (Halbach array) 1.5	Gd (spheres 0.3mm) + 80% wat - 20% glycol
	2014 PM-II	Rotary	2.5	33	Permanent (3 nested Halbach array) 1.5	Gd (spheres 500 μ m) + 80% wat - 20% glycol
Technical University of Denmark, Denmark (Engelbrecht et al., 2011; Eriksen et al., 2014; 2015)	2011	Reciprocating	-	10.2	Permanent (Halbach array) 1.03	Gd; La(Fe,Co,Si) ₁₃ ; (La,Ca,Sr)MnO ₃ (plates) + 75%wat - 25%automotive antifreeze mixture
	2014 MAGGIE	Rotary	138	20	Permanent (Halbach array) 1.4	Gd (spheres 0.3÷0.6 mm); Gd _{100-x} Y _x + 95% demineralized water - 5% glycol
Technical University of Denmark, Denmark and University of Santa Catarina, Brazil (Engelbrecht et al., 2012)	2012	Rotary	1000	25.4	Permanent (Concentric Halbach array cylinders) 1.24	Gd (spheres 0.25÷0.8 mm)
Technical University of Denmark, Denmark + ENOVHEAT partners (Johra et al., 2019)	2018 MagQueen	Rotary	1500 (heat)	25	Permanent 1.6	La(Fe,Mn,Si) ₁₃ H ₂ (spheres)
University of Santa Catarina, Brazil (Trevizoli et al., 2011; 2016; Lozano et al., 2015)	2011	Reciprocating	3.9	4.45	Permanent (Halbach array) 1.65	Gd (plates 0.85 mm) + water
	2015	Rotary	150	12	Permanent 1.0	Gd (spheres 425÷600 μ m) + 80% distilled water - 20%automotive antifreeze mixture
	2016	Rotary	53.7	30	Permanent (nested Halbach array cylinder) 1.69	Gd (spheres 0.55 mm) + 80% water - 20% glycol
University of Ljubljana, Slovenia (Tušek et al., 2013; Tomc et al., 2018)	2012	Reciprocating	42	20	Permanent 1.15	Gd (6 different geometries) + 67% distilled water – 33% ethylene glycol
	2018	Reciprocating	-	13	Electromagnets 1.0	Gd (spheres 0.1÷0.3 mm); + 70% distilled water + 30% glycol
Oak Ridge National Lab, USA (Shassere et al., 2012)	2012	Reciprocating	8.8 * kg ⁻¹	5.6	Electromagnets 0.75	Gd (sheets 1 mm) + water – ethylene glycol
Kobe University, Japan (Kitanovski et al., 2015c)	2011	Rotary	10	5	Permanent 0.6	Gd (packed bed)
Kobe University, Japan Railway Technical Research Institute (Kitanovski et al., 2015c)	2011	Rotary	150	7.6	Permanent (Halbach array)	Gd; La-Fe-Co-Si (packed bed)

Kobe University, Japan + Railway Technical Research Institute + Companies and other universities Japan (Hirano et al. 2014)	2014	Rotary	1400	21	Permanent (Halbach array) 1.5	Gd; Gd-Y; Gd-Dy (0.5 - 0.85 mm) (flat plates)
Sanden Co. Inc., Japan (Kitanowski et al., 2015c)	2013	Rotary	200	13.8	Permanent (Halbach array) 0.85	Gd; Gd alloy (packed bed)
University of Genoa, Italy (Tagliafico et al., 2013)	2013	Reciprocating	-	5	Permanent 1.5	Gd (parallel-plates 0.8 mm) + 50% water-50% ethanol
University of Salerno; University of Naples Federico II, Italy + University of Victoria, Canada (Aprea et al., 2014; 2015; 2016a; 2016b)	2014 8 Mag	Rotary	200	13.5	Permanent (Halbach array) 1.25	Gd (spheres 400÷500 µm) + distilled water
University of Bologna; IMEM-CNR in FRIMAG project, Italy (Rosati et al., 2018)	2018	Rotary	-	-	Permanent (Halbach array) 1.4	Gd (parallel-plates 0.8 mm) + water
University of Coruna, Spain (Gómez et al., 2013b)	2013	Reciprocating	16.7 * kg ⁻¹	3.5	Permanent 1.0	Gd (parallel-plates 0.5 mm) + distilled water
University of Zaragoza, Spain (Velázquez et al., 2014)	2014	Reciprocating	6	19.3	Permanent 1.4	Gd (spheres 0.2 – 0.4 mm) + 75% water - 25% glycol
Astronautics Corporation of America, USA. (Jacobs et al., 2014)	2014	Rotary	3042	18	Permanent 1.44	LaFeSiH alloys + water
Wroclaw University of Technology (Czernuszewicz et al., 2014a)	2014	Reciprocating	-	6.9	Permanent (Halbach array) 1.0	Gd (6 different geometries) + water - ethane-1,2-diol mixture
TU Delft, Netherlands with South China University of Technology, China (Huang et al., 2016; 2018)	2016	Rotary	40	9.1	2 Permanent N-shaped structure	Gd (spheres 0.4 – 0.8 mm)
KTH Royal Institute of Technology, Sweden (Monfared & Palm, 2016)	2016	Rotary	-	18	Permanent 1.1	La-Fe-Mn-Si-H (packed bed irregular-shape particles)
Baotou Research Institute of Rare-earths, China (Zhang et al., 2016)	2016	Rotary	-	25	Permanent concentric Halbach array 1.5	Gd; GdEr alloy (particles) + demineralized water
TU Darmstadt, Germany (Benke et al., 2018)	2018	Reciprocating	-	-	Permanent	La-Fe-Si alloys
TOBB University of Economics & Technology; ASELSAN; Ankara University, Turkey (Baser et al., 2018a; 2018b)	2018 MAGICE -1	Reciprocating	75	12	Permanent Halbach array	Gd spheres (0.5 mm) + 80%water-20%ethylene glycol
	2018	Rotary	500	5	Permanent Four-pole	Packed bed
Cooltech Application Company, (Kitanovski et al., 2015c; Chaudron et al., 2018; Lionte et al., 2018b)	2011	Reciprocating	150	38	Permanent 1.27	Gd; Gd-Tb (parallel-plates)
	2013	Rotary	120	42	Permanent 0.98	Gd; Gd-Tb (packed-bed)
	2014	Rotary	300	38	Permanent 1.17	Gd; Gd-Er/La-Fe-Si (parallel-plates)

	2018	Rotary	-	30	Permanent 1	LaFeSi alloys (parallel-plates)
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Table 3. Electrocaloric devices built to date and research group, listed with their general characteristics.

Research group and location	Year Name	Scale	Application AER/TD	$Q_{\text{ref,max}}$ [W]	$\Delta T_{\text{span,max}}$ [K]	ECM + HTF/TD ΔE_{max} [MJ m ⁻¹]
National Bureau of Standards, Corning Glass Works, USA (Radebaugh et al., 1979)	1979	Macroscale	Cryogenic TD	-	0.3	SrTiO ₃ ceramics (parallel-plates) + magnetothermal-mechanical heat switches 2
Moscow Power Engineering Institute, URSS (Sinyavsky et al., 1989; Sinyavsky & Brodyansky, 1992)	1989	Macroscale	Room – temperature AER	-	5	multilayer of PbSc _{0.5} Ta _{0.5} O ₃ (PST) + Liquid pentane 6
CeramPhysics, Columbus, OH 43081, USA (Lawless, 2005)	2005	Macroscale	Room - temperature AER	-	-	multilayer of x(PbMg _{0.33} Nb _{0.67} O ₃)+ (PbSc _{0.5} Nb _{0.5} O ₃) or x(PbMg _{0.33} Nb _{0.67} O ₃)+ (PbTiO ₃)+z(SrTiO ₃)
Pennsylvania State University, USA (Gu et al., 2013; Zhang et al., 2017)	2013	Macroscale	Room – temperature TD	-	6	Poly (vinylidene fluoridetrifluoro-ethylene) 68/32 mol% (ei-copolymer) 100
	2017	Macroscale	Room – temperature TD	-	2	commercial Y5V ceramic composition 16.6
University of Ljubljana, Slovenia (Plaznik et al., 2014; 2015; 2019)	2014	Small-Scale	Room – temperature AER	-	3.3	PMN-10PT (bulk ceramic plates) + silicone oil 5
	2019	Small-Scale	Room – temperature AER	16* kg ⁻¹	3.3	PMN-10PT (bulk ceramic plates) + silicone oil 5.7
Leibniz Universität Hannover, Fraunhofer Institute for Ceramic Technologies and Systems, Germany (Blumenthal et al., 2016)	2016	Small-scale	Room – temperature AER	-	0.17	PMN-PT (parallel plates 0.8) + silicone oil 1
Parc, a Xerox, USA (Wang et al., 2018)	2018	Small-scale	Room – temperature TD	-	2.5	BaTiO ₃ multilayer capacitors (1.9 mm) 30.8

Table 4. Elastocaloric devices built to date and research group, listed with their general characteristics.

Research group and location	Year Name	Driving mechanisms $\epsilon_{\max}$ [%]	$\dot{Q}_{\text{ref,max}}$ [W]	$\Delta T_{\text{span,max}}$ [K]	eCM + HTF
University of Maryland; Pacific Northwest National Lab, USA (Cui et al., 2012)	2012	Rotating 8.5	-	17	NiTi wires (0.5 mm diameter) + air
Saarland University, Germany (Schmidt et al., 2015)	2015	Two linear drivers 4	-	7	Ni-Ti alloy + no HTF (directly eCM)
University of Maryland; Niron Magnetics Inc., USA (Qian et al. 2015a; 2015b)	2015	Linear 4.5	65	4.7	NiTi tubes (0.5 mm diameter) + water
Technical University of Denmark, Denmark; University of Ljubljana, Slovenia (Tušek et al., 2016; Engelbrecht et al., 2017)	2016	Linear 4	782 kg^{-1}	19.9	Ni-Ti plates (tck=0.35mm) + water